

BioNetGen Overview

Jim Faeder

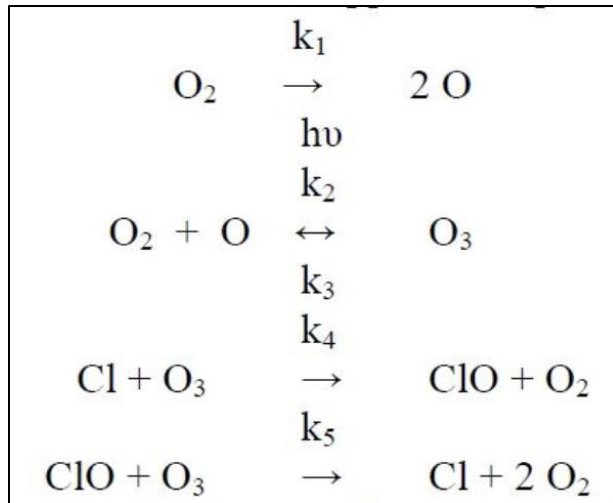
*MSCBIO 2025, Introduction to
Bioinformatics Programming in Python*

September 17, 2024

Reaction Networks Models can be used to describe a wide range of natural processes.

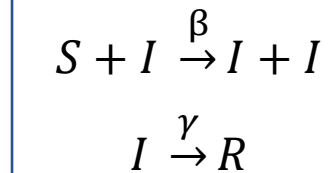
Chemistry

Ozone formation in the upper atmosphere



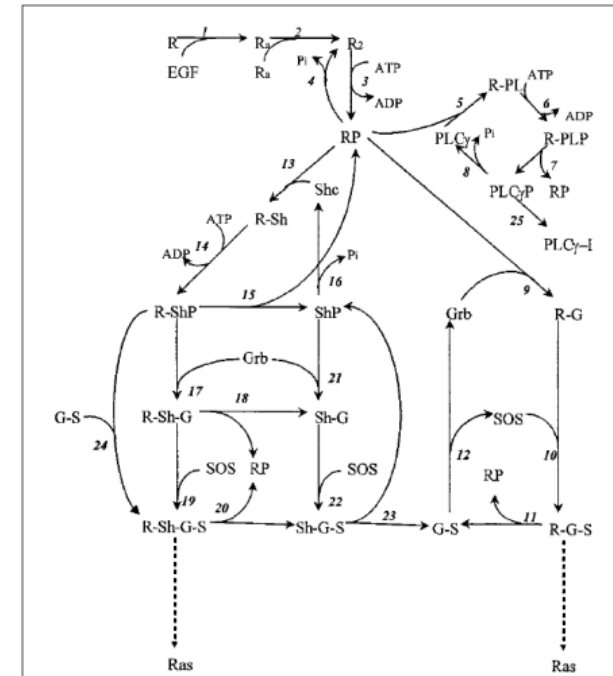
Epidemiology

Susceptible-Infected-Recovered (SIR)

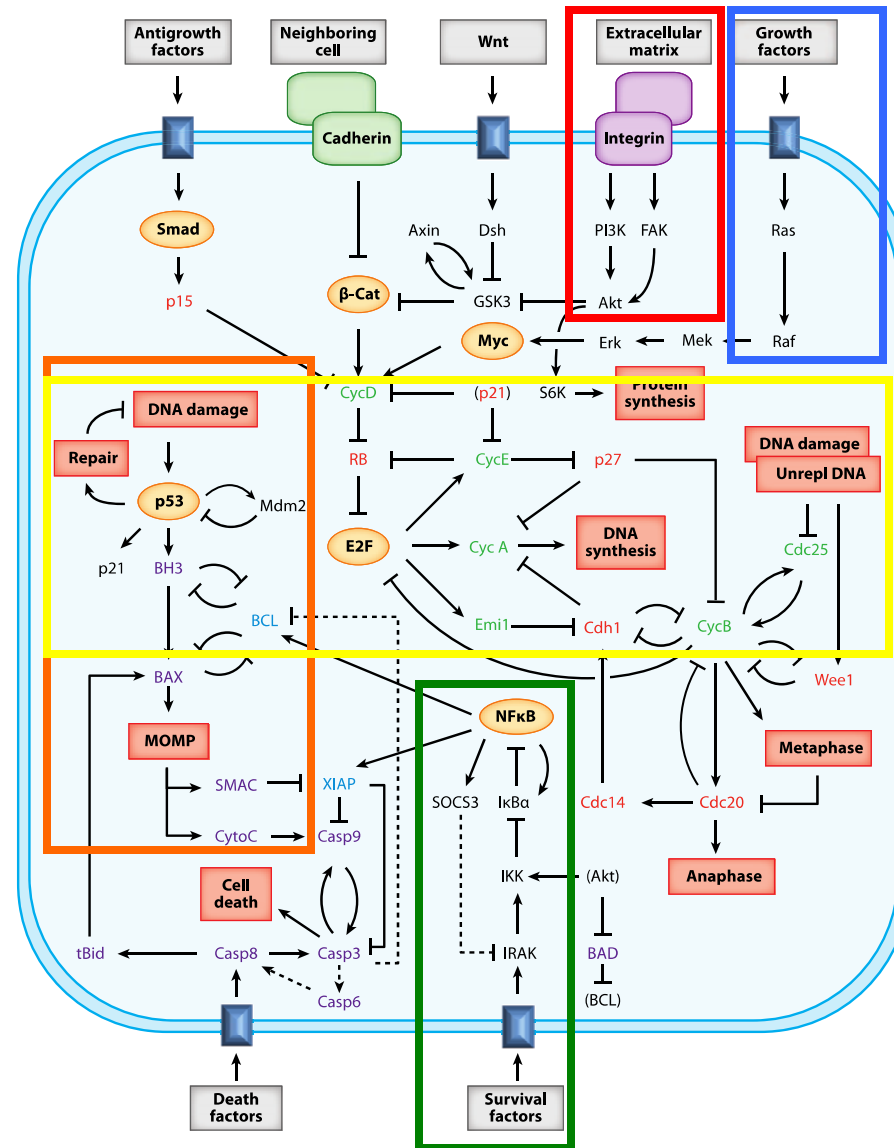


Molecular biology

Signaling network

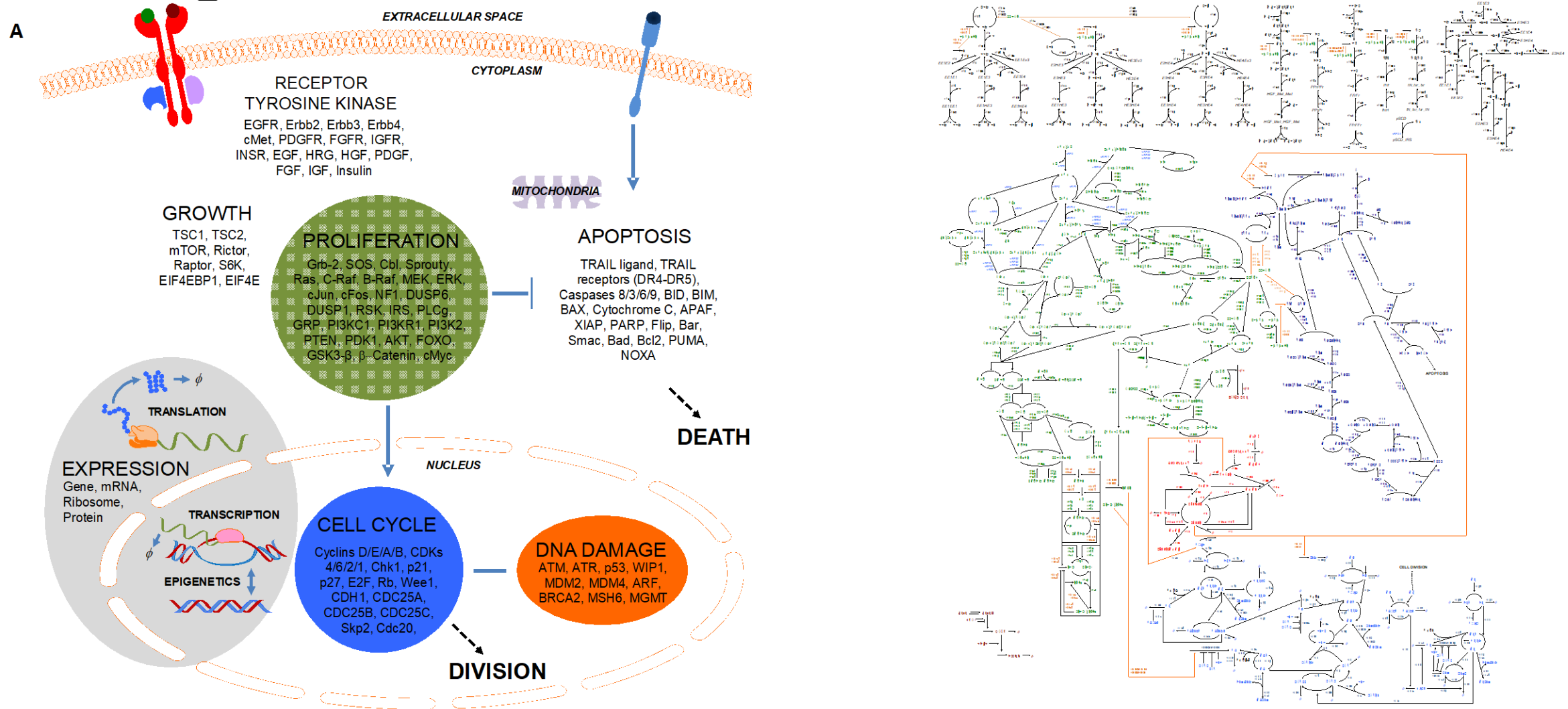


Toward an integrated model of the cell



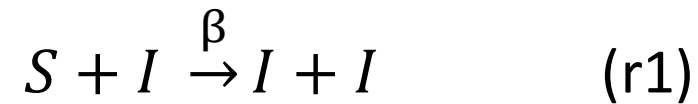
How to accomplish the task of linking together signaling modules that determine cell fate?

Integrated Model of MCF10A Cell



Reaction kinetics are defined by a few simple concepts.

Consider the **reaction**



The entities S and I are **species**, which represent pools of identical (i.e., indistinguishable) entities. Here S and I represent the numbers of susceptible infected individuals respectively. The reaction describes how a susceptible individual becomes infected by an infected individual.

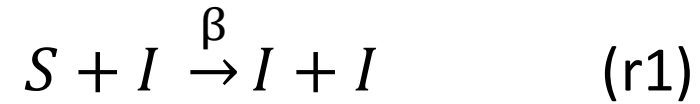
The parameter β is a **rate constant**, and the rate at which the reaction occurs is given by

$$r_1 = \beta \cdot S \cdot I,$$

according to the simplest description given by **mass action kinetics**, which says that the rate of a reaction is given as the product of the reactant concentrations times the **rate constant**.

Reaction network models (RNM) are built from individual reactions.

The basic SIR model contains two **reactions**



and three **species**: S , I , and R , which represents the number of recovered individuals.

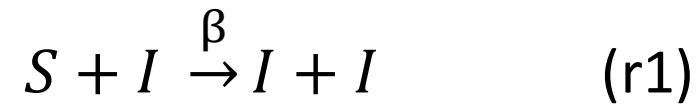
An RNM tracks the values of the species concentrations as a function of time, which can be written as the vector

$$\mathbf{X}(t) = \begin{pmatrix} S \\ I \\ R \end{pmatrix}$$

A complete RNM consists of a set of **reactions**, **species**, **rate constants**, and the **initial values** (also called **initial concentrations**) of the species, denoted $\mathbf{X}(0)$.

Ordinary differential equations (ODEs) describe the mean behavior of a reaction network.

The SIR model contains two **reactions**



which give rise to three ODE's

$$\begin{array}{l} \frac{dS}{dt} = \\ \frac{dI}{dt} = \\ \frac{dR}{dt} = \end{array} \begin{array}{|c|c|} \hline \text{r1} & \text{r2} \\ \hline -r_1 & \\ +r_1 & -r_2 \\ & +r_2 \\ \hline \end{array}$$

$$\begin{array}{l} \frac{dS}{dt} = \\ \frac{dI}{dt} = \\ \frac{dR}{dt} = \end{array} \begin{array}{l} -\beta \cdot S \cdot I \\ \beta \cdot S \cdot I - \gamma \cdot I \\ \gamma \cdot I \end{array}$$

where the rates are given by

$$\begin{array}{l} r_1 = \beta \cdot S \cdot I \\ r_2 = \gamma \cdot I \end{array}$$

The ODEs for RNMs have a compact representation using the stoichiometry matrix.

$$\frac{d\mathbf{X}(t)}{dt} = \begin{bmatrix} \frac{dS}{dt} = -r_1 \\ \frac{dI}{dt} = +r_1 \\ \frac{dR}{dt} = -r_2 \end{bmatrix} = \begin{pmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$$

S • **v(X)**

Stoichiometry matrix Reaction fluxes

The elements of **S** are given by

$$S_{ij} = \begin{aligned} &(\# \text{ times species } i \text{ appears as product in reaction } j) \\ &-(\# \text{ times species } i \text{ appears as reactant in reaction } j) \end{aligned}$$

ODEs for most RNMs are solved numerically.

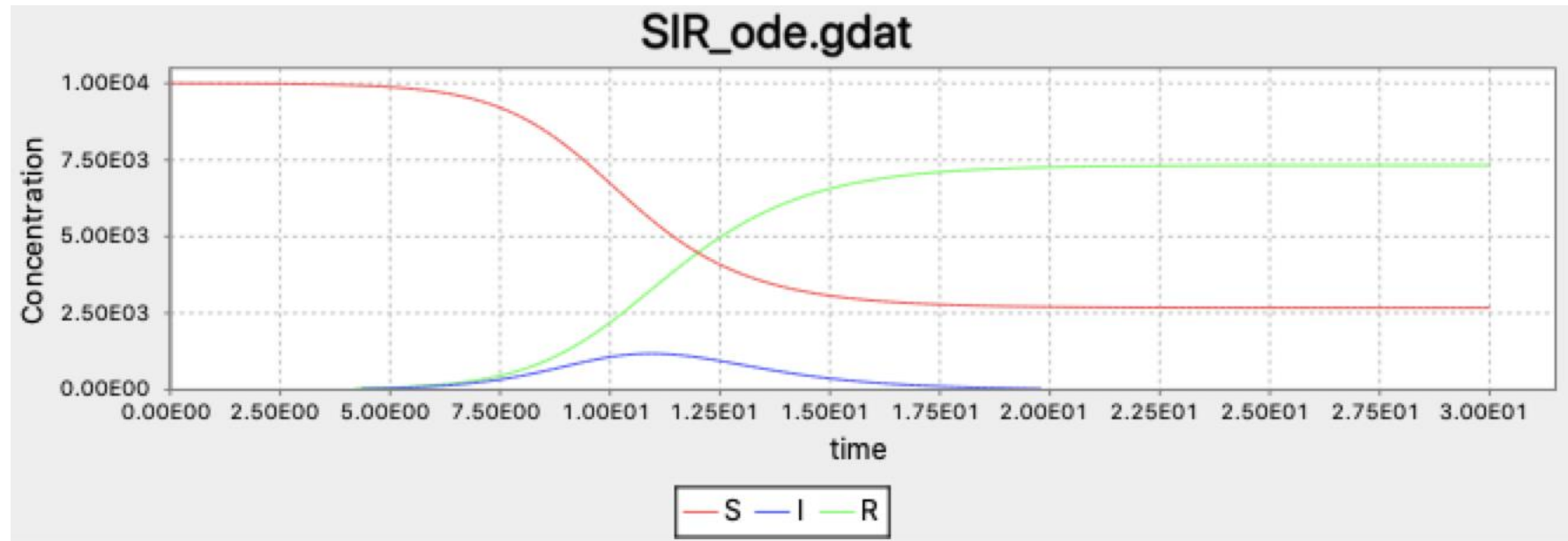
The general equation for RNM dynamics is

$$\frac{d\mathbf{X}(t)}{dt} = \mathbf{S} \cdot \mathbf{v}(\mathbf{X})$$

- These equations are in general *nonlinear*.
- They cannot be solved analytically except in special cases.
- Numerical integration is required to obtain a solution.
- “Stiffness” arises because reactions occur on different timescales, requiring special techniques to obtain accurate solutions efficiently.
- Widely used solvers are CVODE and LSODA packages.
- Accessed using ode method in the simulate command of BNG.

Numerical solution of SIR model describes dynamics of infection outbreak.

Time course for S , I , and R produced by BioNetGen and displayed in RuleBender.



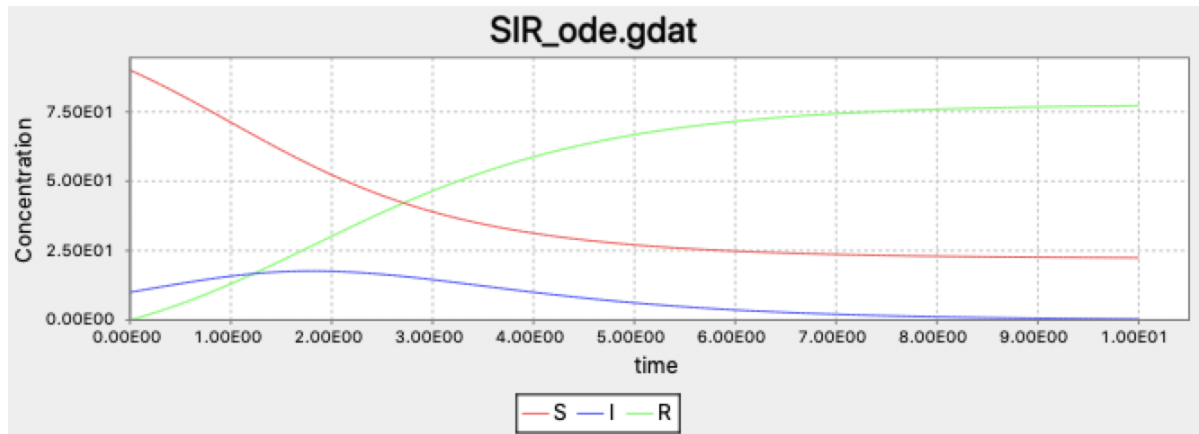
$$\begin{pmatrix} \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 1.8 \times 10^{-3} \\ 1 \end{pmatrix}, \quad \begin{pmatrix} S(0) \\ I(0) \\ R(0) \end{pmatrix} = \begin{pmatrix} 9,999 \\ 1 \\ 0 \end{pmatrix}$$

ODEs ignore the discrete nature of individuals in a network and neglect the effects of noise.

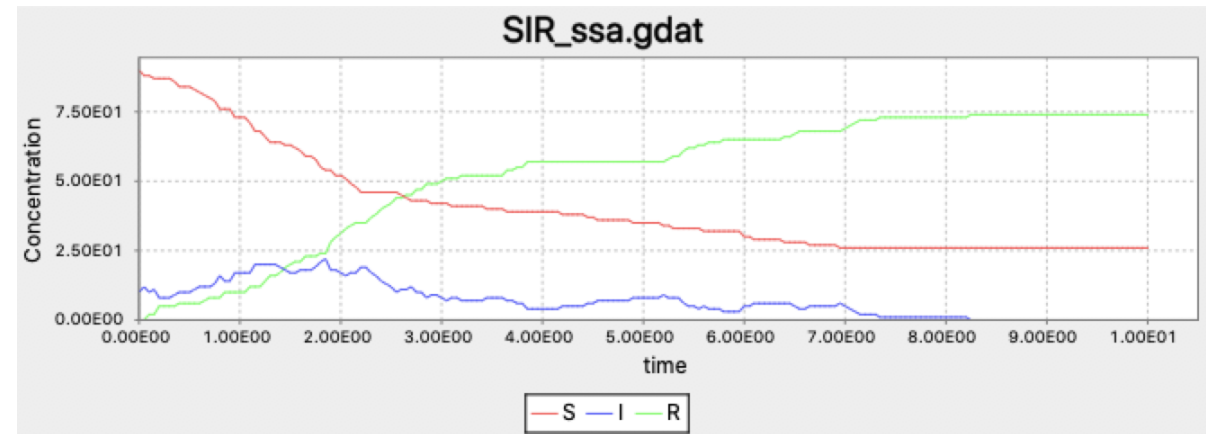
- ODEs treat all variables as continuous and cannot account for the discrete nature of individuals.
- ODEs in general (with exceptions) describe the *average* behavior of a system, but noise can affect real-world dynamics, e.g., disease outbreaks.
- The same reaction network description can be used to generate stochastic time courses (“trajectories”) that exactly sample the true probability distribution.

Stochastic simulations can reveal the effects of noise in finite populations.

ODE simulation



Stochastic simulation



SIR model with

$$\begin{pmatrix} \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 1.8 \times 10^{-2} \\ 1 \end{pmatrix}, \quad \begin{pmatrix} S(0) \\ I(0) \\ R(0) \end{pmatrix} = \begin{pmatrix} 90 \\ 10 \\ 0 \end{pmatrix}$$

Stochastic dynamics can be computed with the Gillespie Algorithm.

BioNetGen's ssa method

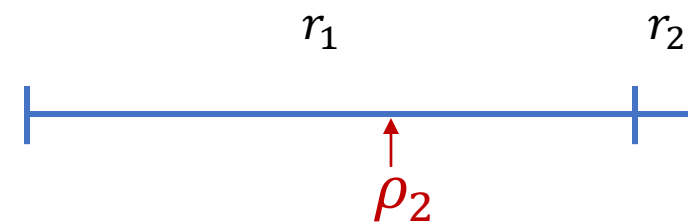
Gillespie's **Direct Method** illustrated for the SIR model. At time $t=0$ we have

$$\mathbf{X}(0) = \begin{pmatrix} 90 \\ 10 \\ 0 \end{pmatrix}, \quad \mathbf{v}(0) = \begin{pmatrix} \beta S \cdot I \\ \gamma I \end{pmatrix} = \begin{pmatrix} 16.2 \\ 1 \end{pmatrix}$$

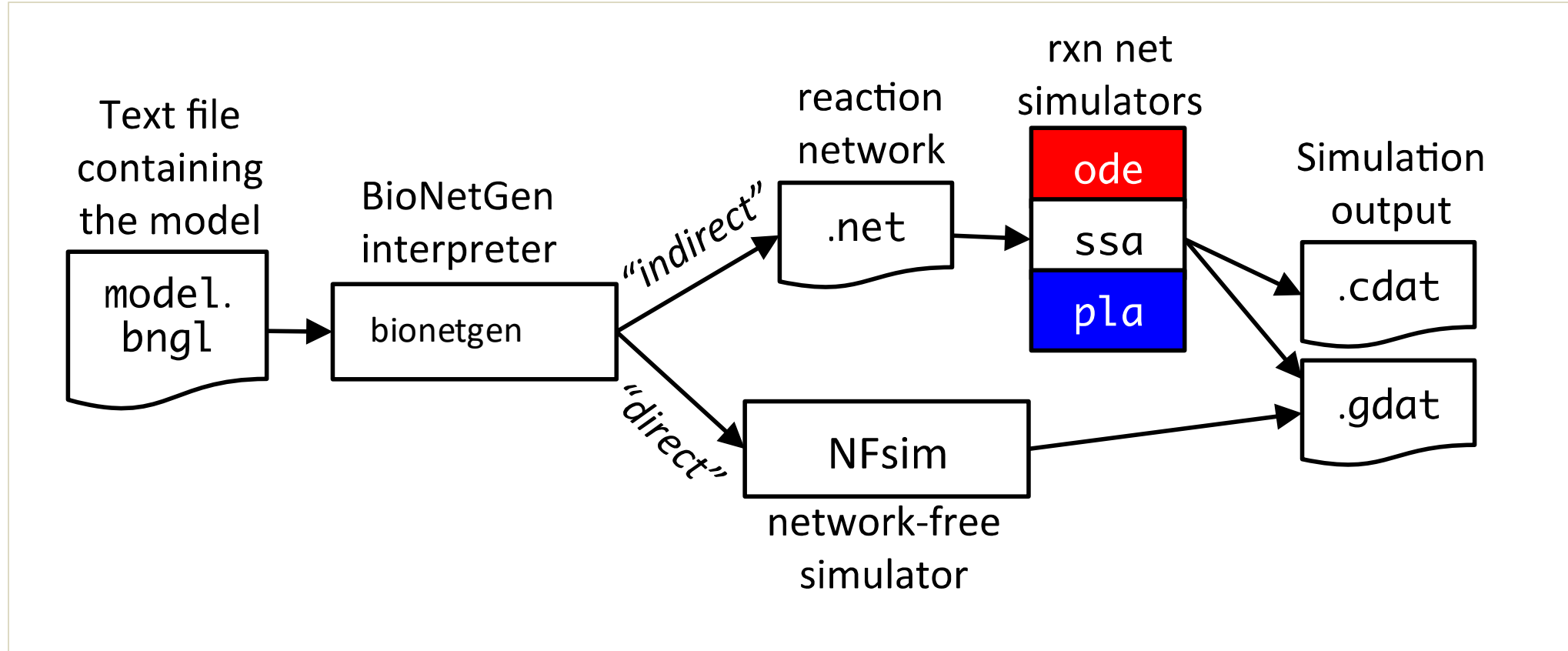
Total rate of all reactions, a_0 , is 17.2.

1. Time of next reaction is chosen as $\tau = \frac{-\ln \rho_1}{a_0}$, $t = t + \tau$
2. Index of next reaction, j , is chosen using so probability of each reaction firing is proportional to its rate.

$$\mathbf{X}(\tau) = \mathbf{X}(0) + \mathbf{s}_1 \quad \mathbf{s}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$



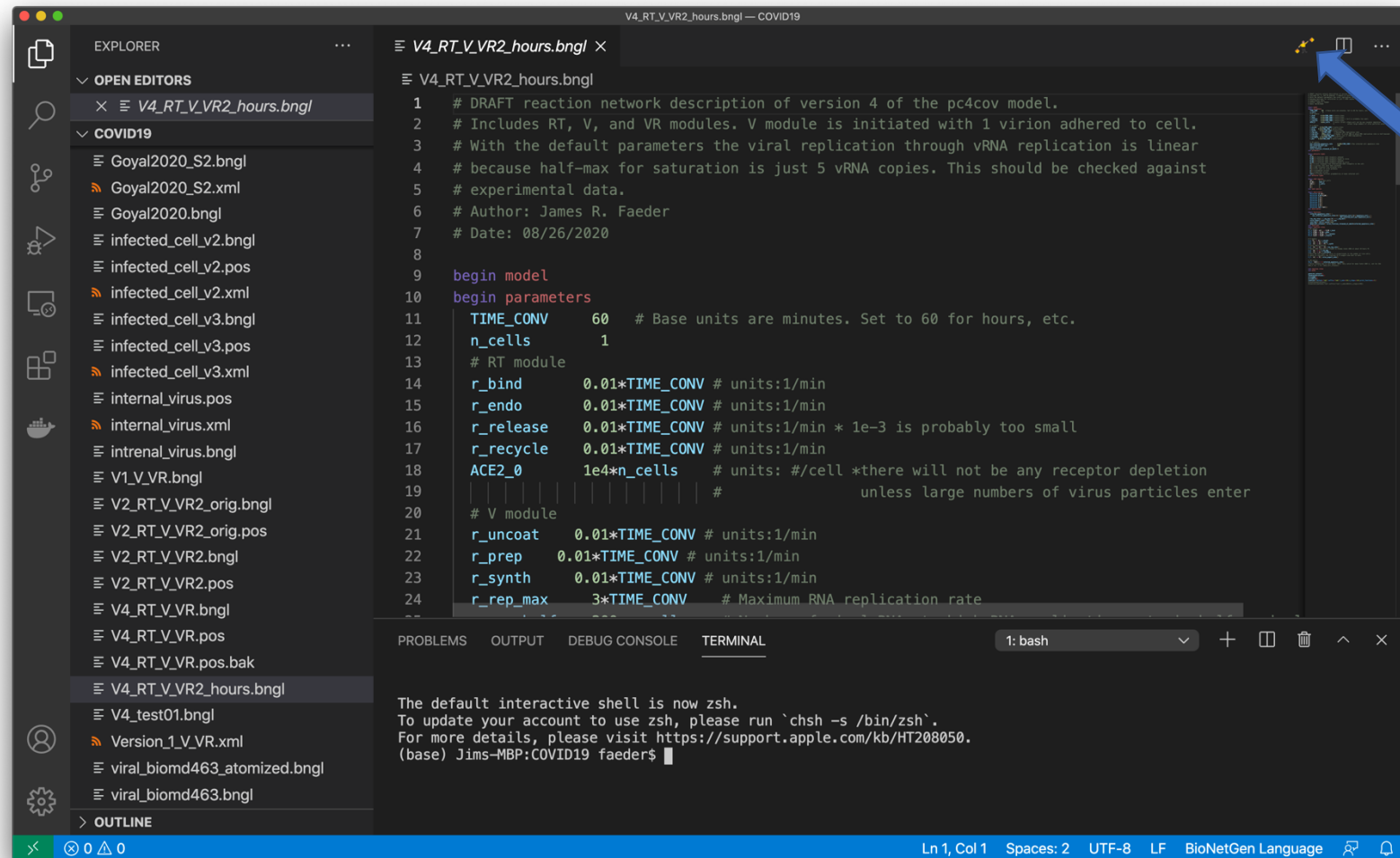
BioNetGen enables the construction, simulation and analysis of RNMs.



Command line interface

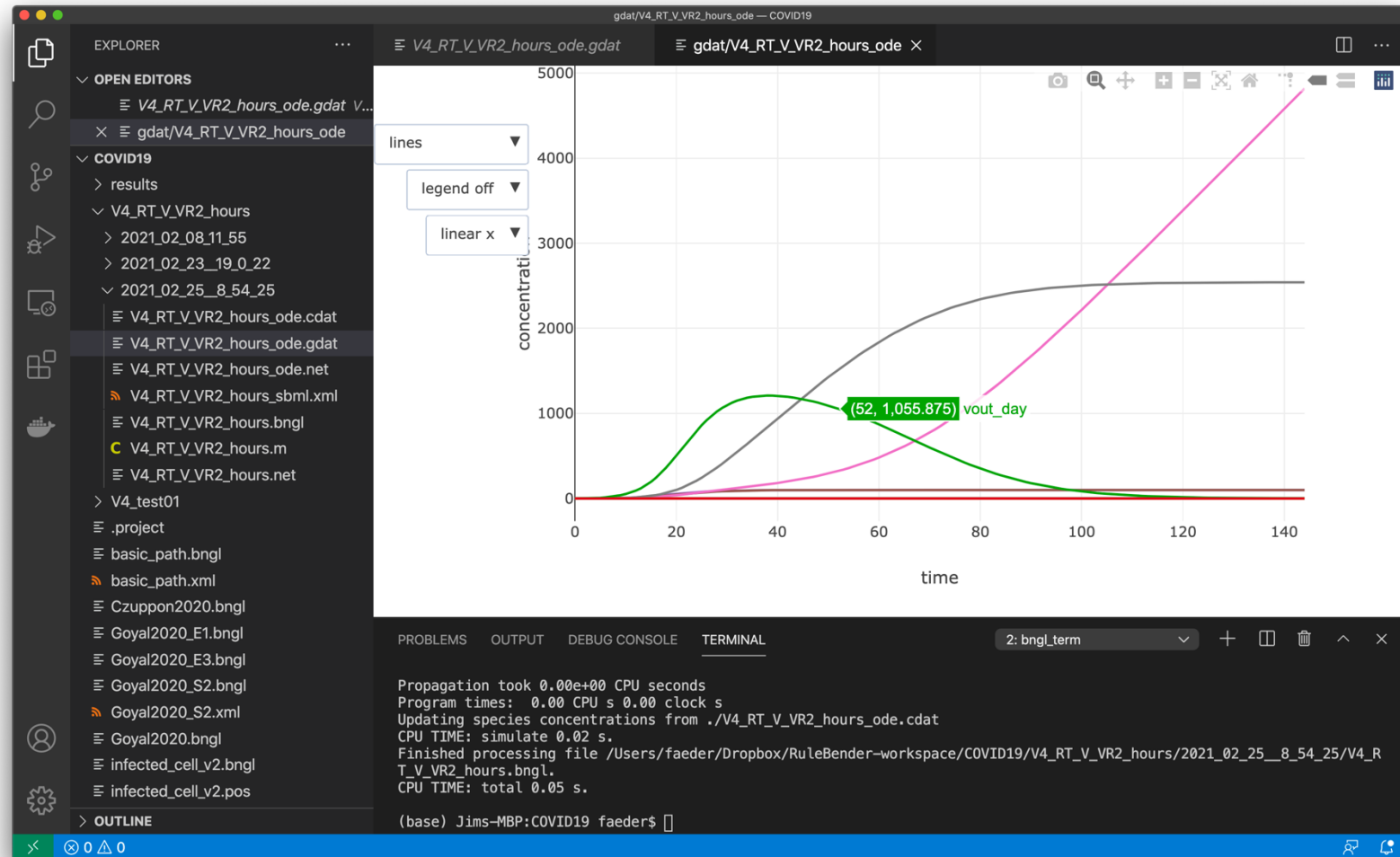
```
>bionetgen run -i model.bngl
```

Visual Studio Code (VS Code) extension provides interactive environment for BioNetGen



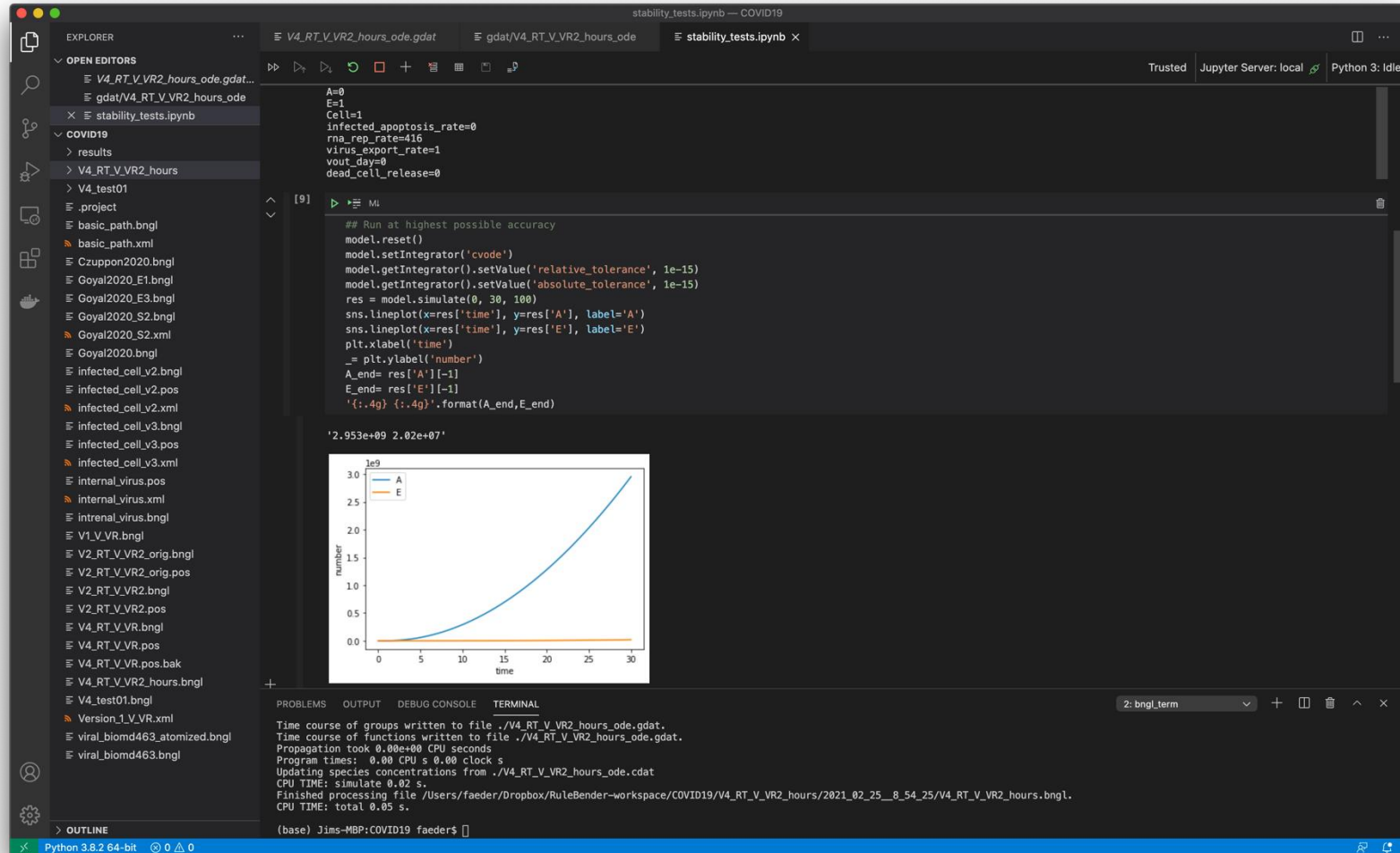
Action button for
model simulation

Visual Studio Code (VS Code) extension provides interactive environment for BioNetGen



Interactive
plotting

Visual Studio Code (VS Code) extension provides interactive environment for BioNetGen

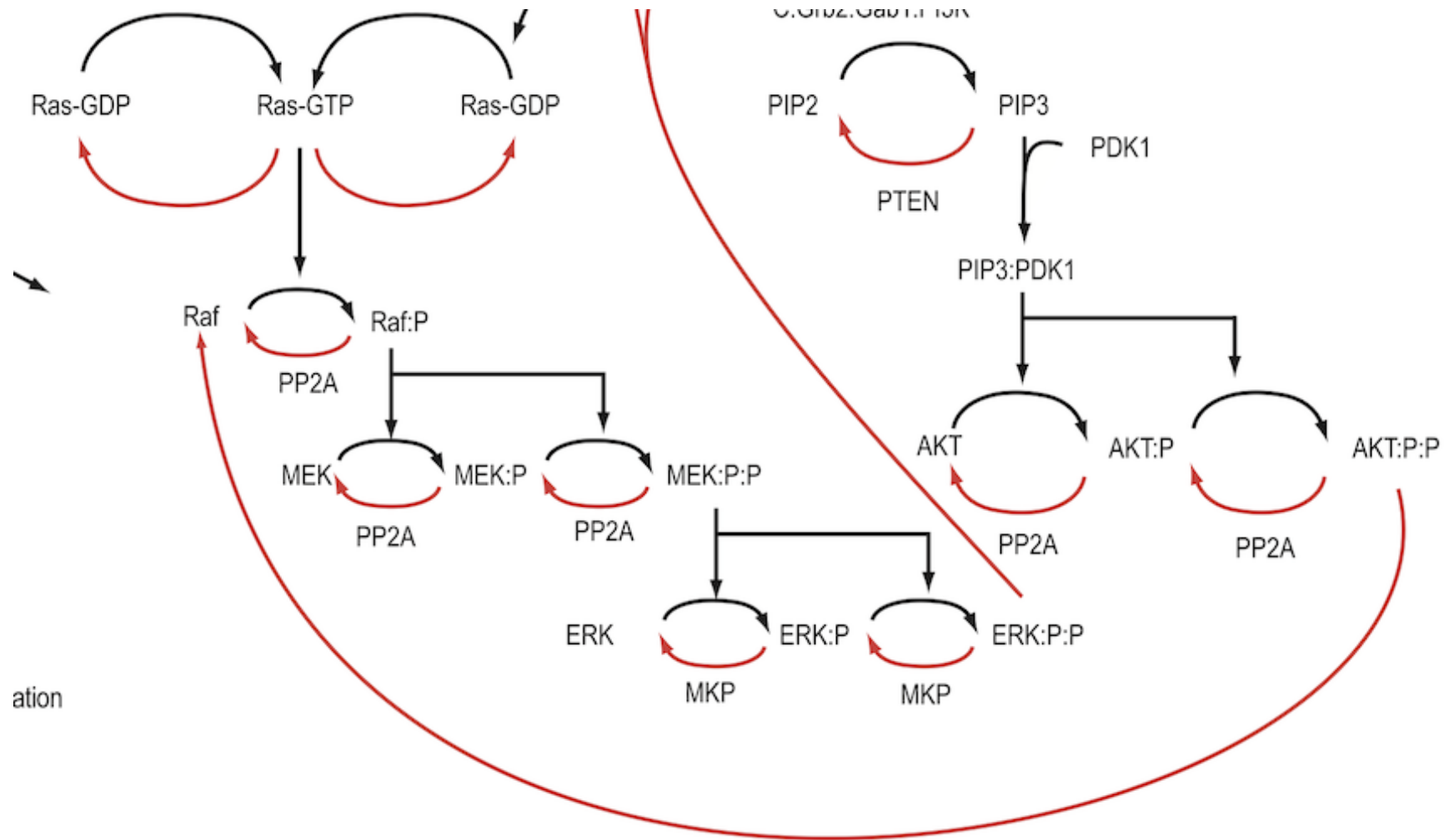


Deeper model
analysis using
Jupyter
notebooks

Stop here

Build SIR model using BioNetGen Extension in VS Code

MAPK model diagram



Overview of Rule-based modeling of cell signaling

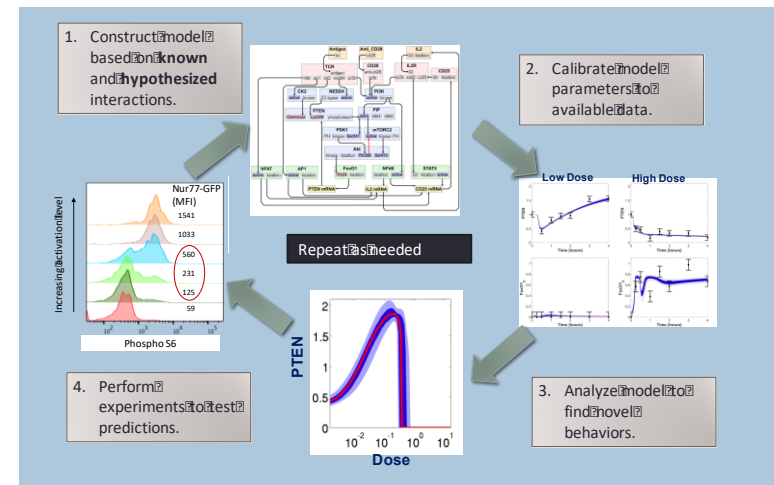
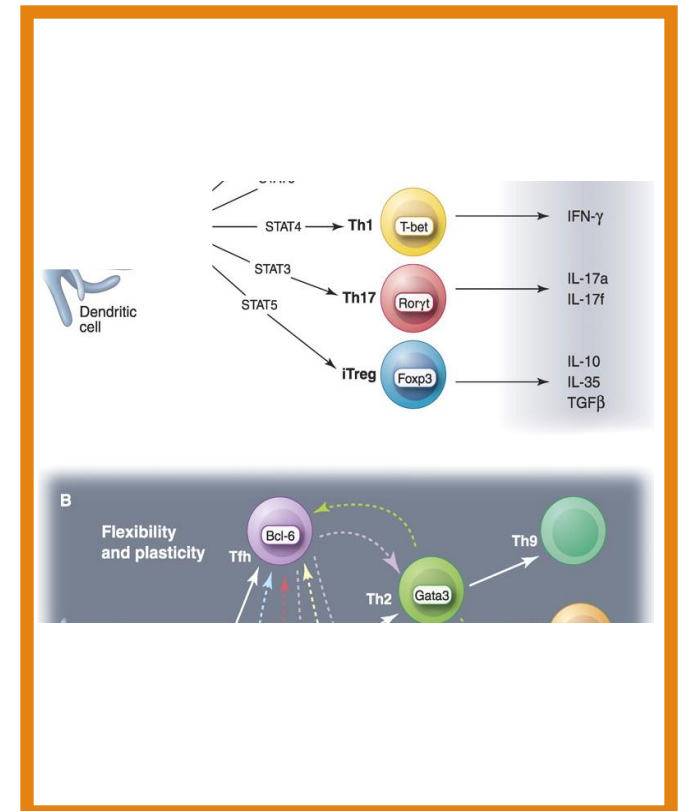
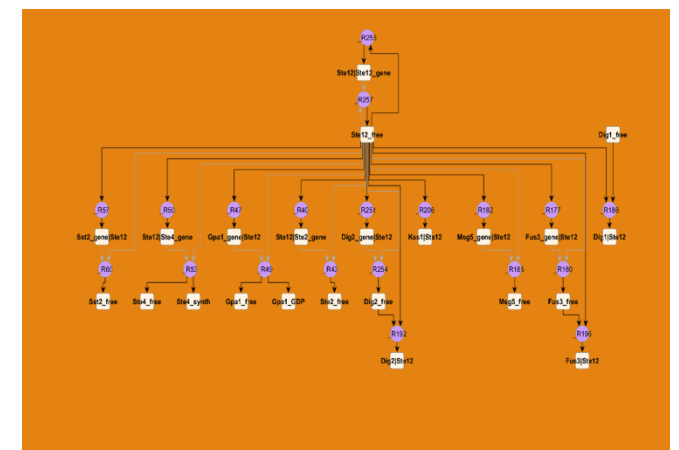
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DEPARTMENT OF COMPUTATIONAL
AND SYSTEMS BIOLOGY

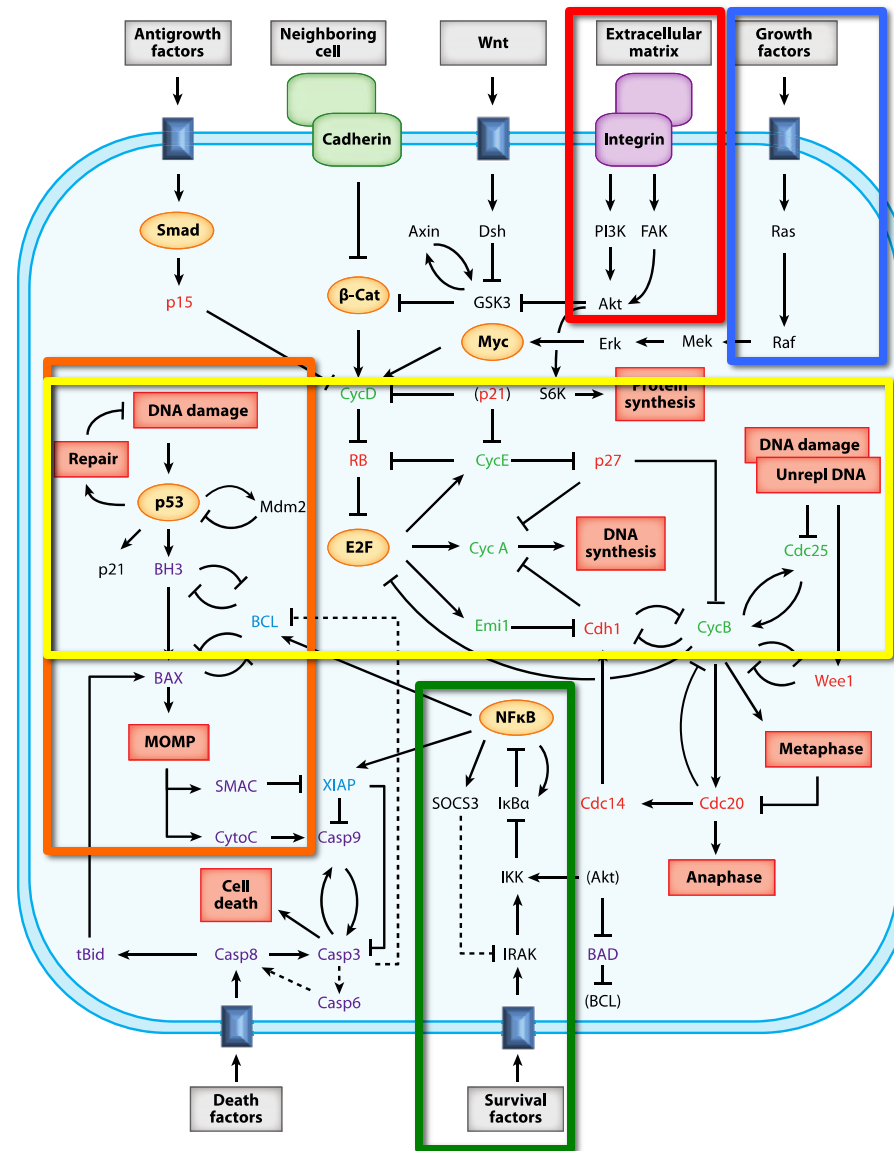
UNIVERSITY OF PITTSBURGH
SCHOOL OF MEDICINE



MMBioS Cell Modeling Workshop
July 12, 2021

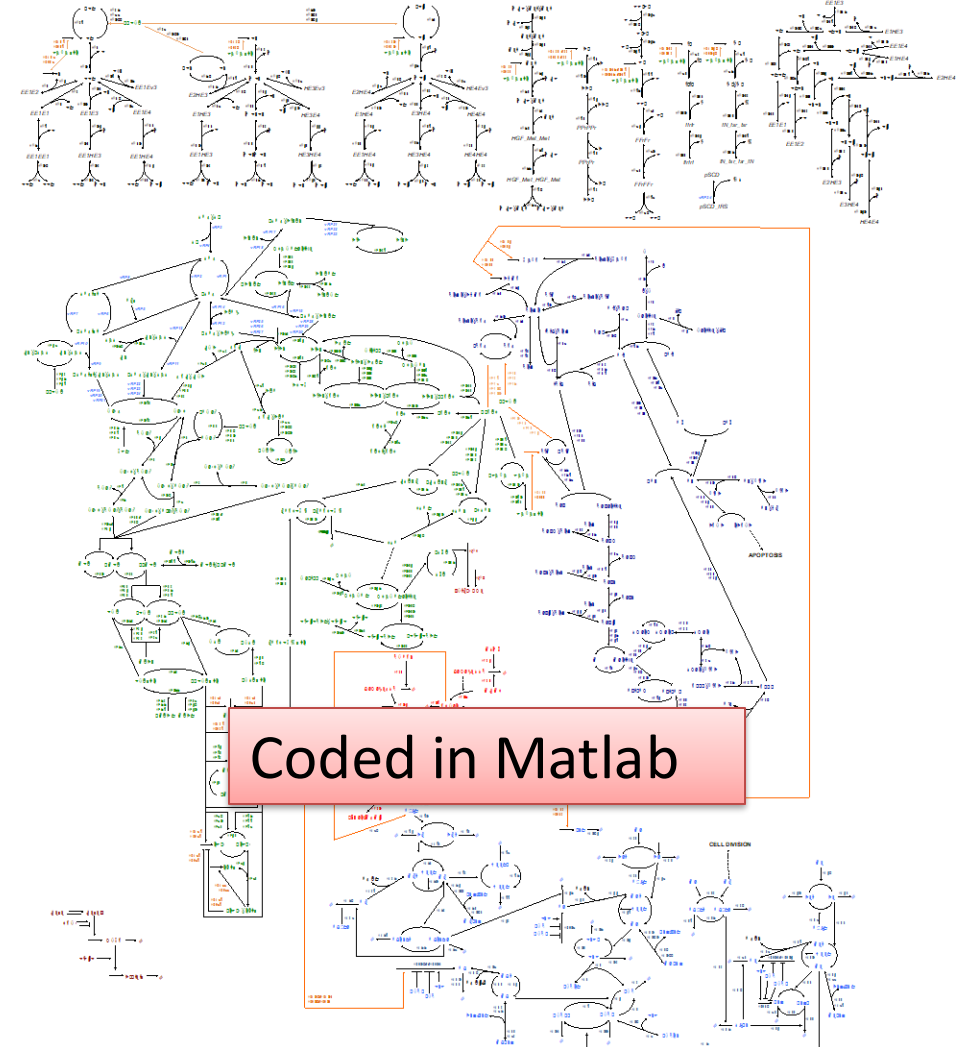
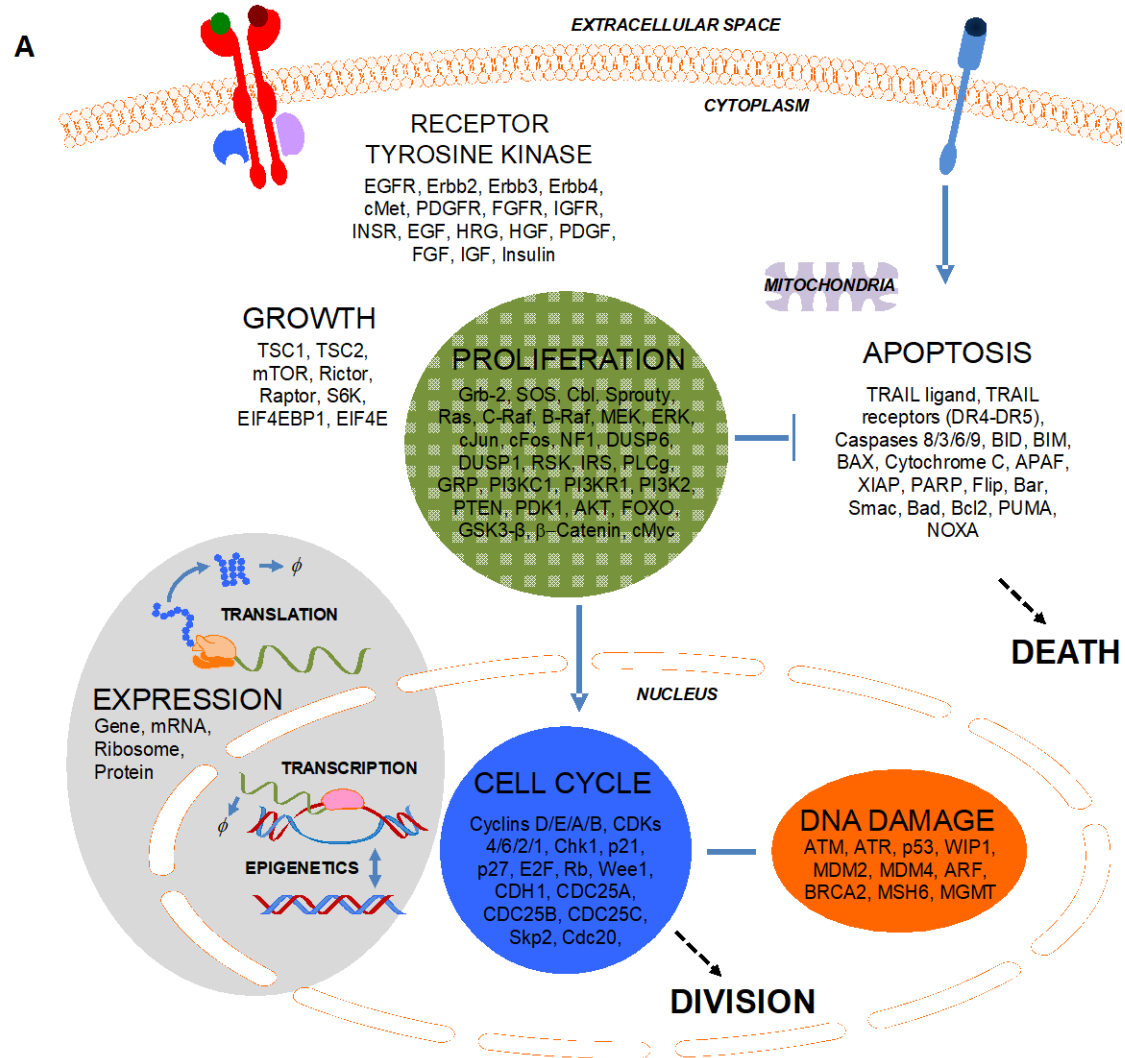


Toward an integrated model of the cell



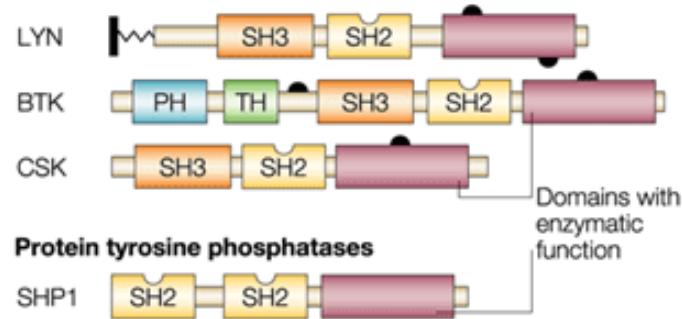
How to accomplish the task of linking together signaling modules that determine cell fate?

Integrated Model of MCF10A Cell



Challenges of modeling cell regulatory networks

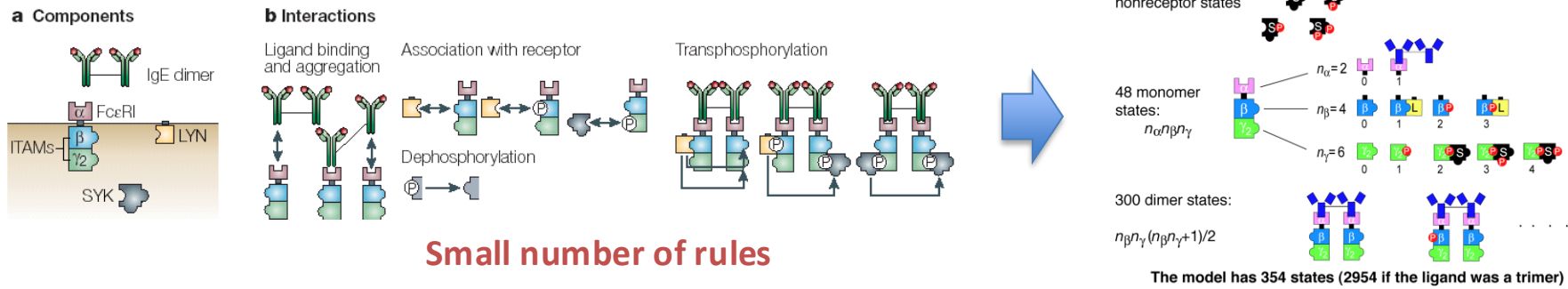
- Proteins are multi-functional



multiple sites of binding

multiple sites of posttranslational modification

- Representing their known interactions requires handling of *combinatorial complexity*



Small number of components and interactions → huge number of possible species and reactions

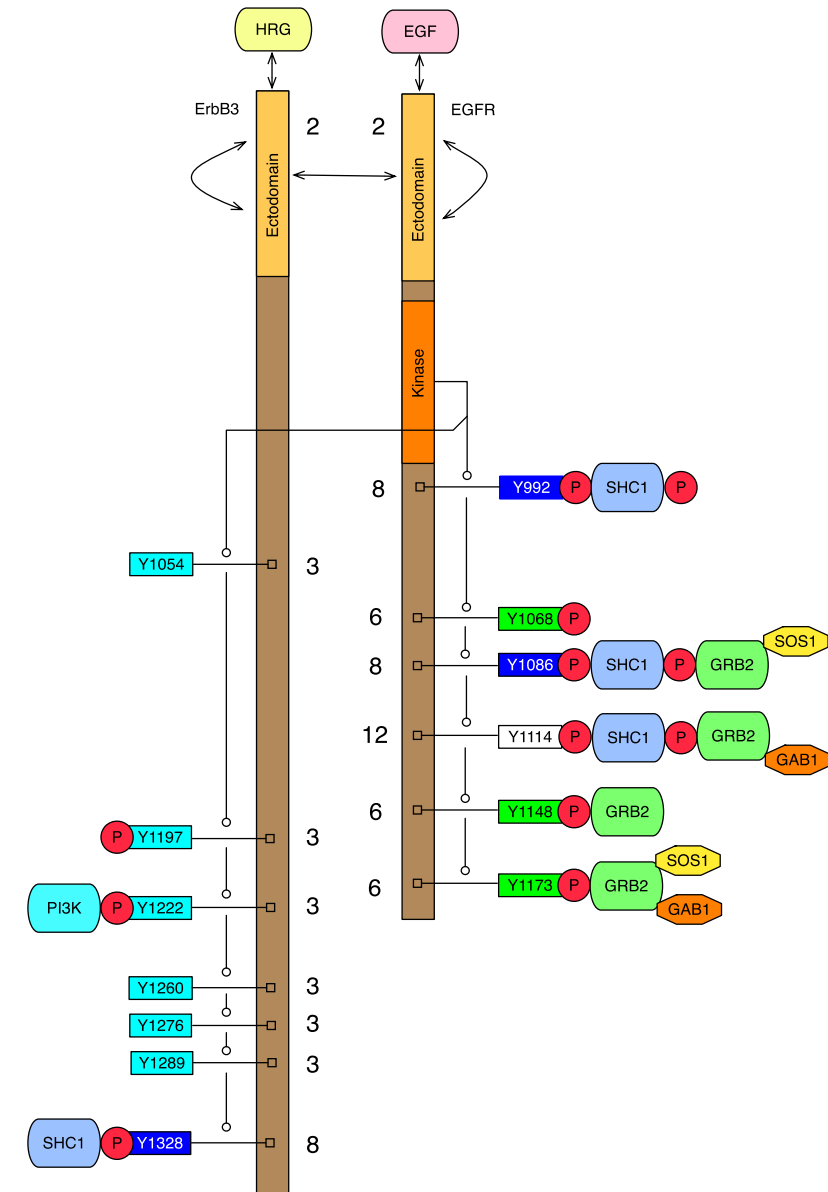
Combinatorial complexity in a realistic model of EGFR signaling

ErbB3:ErbB1 has $> 3.8 \times 10^9$ states

ErbB1:ErbB1 has $> 5.5 \times 10^{10}$ states

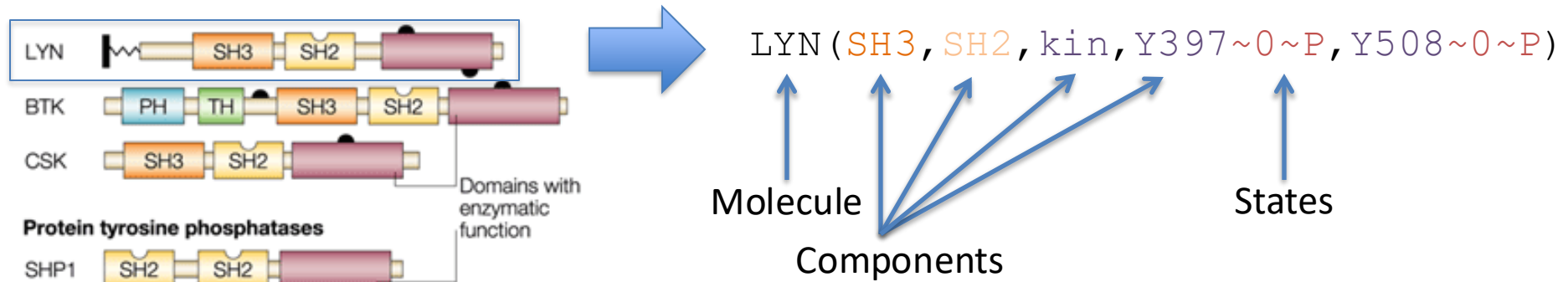
Such models can be built from a manageable number of rules – 10's to 100's

Creamer et al. (2012) *BMC Syst. Biol.*

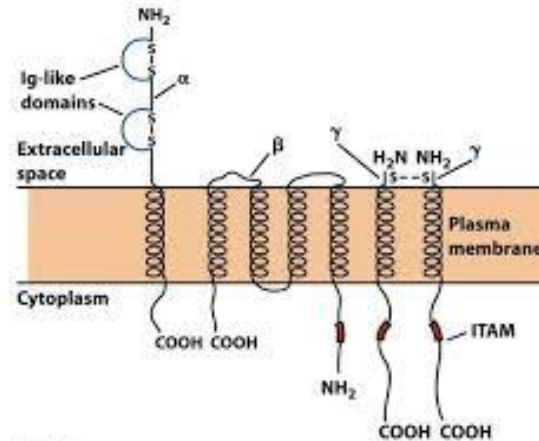


What is Rule-based Modeling (RBM)?

Molecules are modeled as *structured objects*



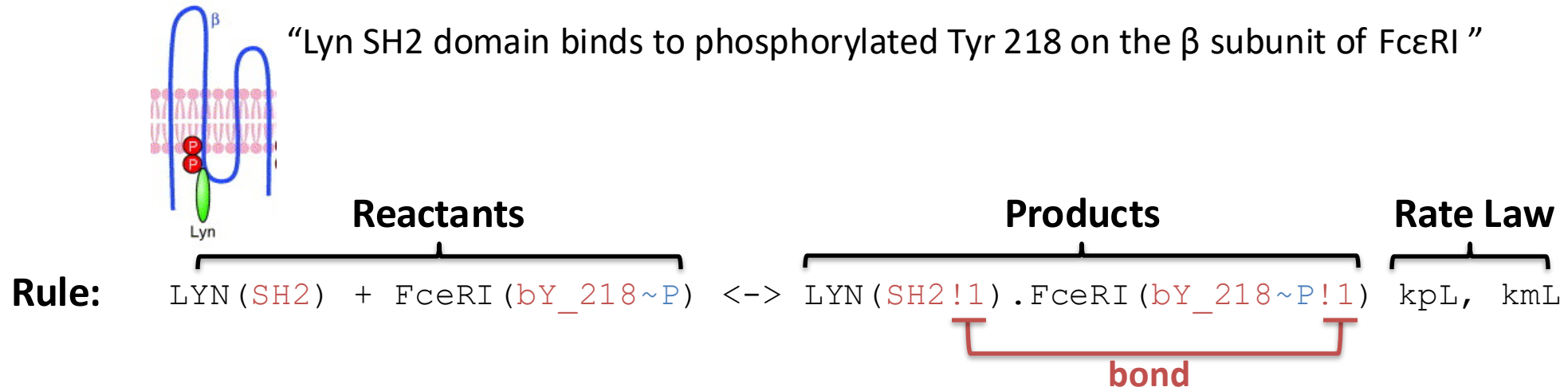
FcεRI: High-affinity IgE receptor



`FcεRI (a_Ig, b_Y218~0~P, g_ITAM~0~P)`

What is Rule-based Modeling (RBM)?

Rules define the interactions of molecules



“Don’t write don’t care” – elements not mentioned may be in any state

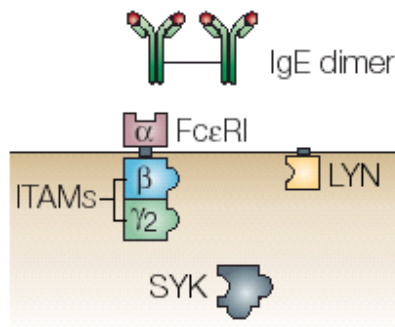
→ *One rule can generate reactions involving many different species*

Reaction rate determined by **Mass Action kinetics**

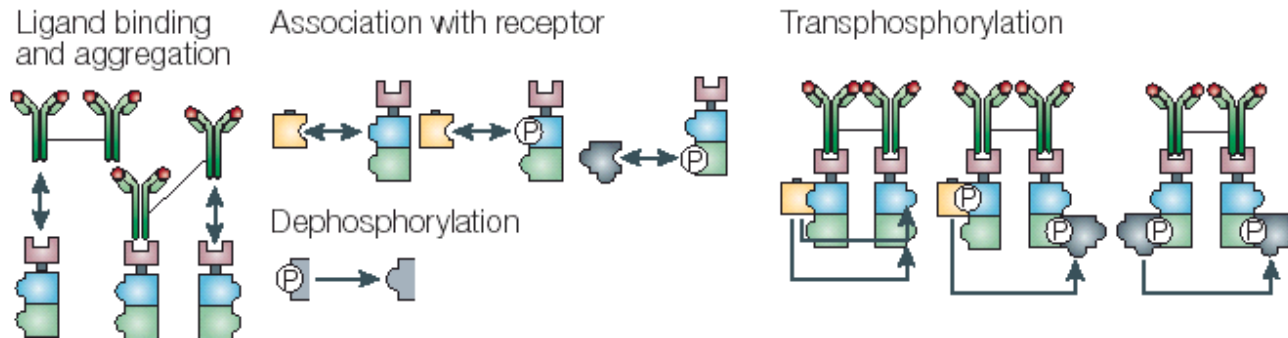
$$\text{rate forward} = k_{\text{pL}} * [\text{Lyn}(\text{SH2})] * [\text{Fc}\epsilon\text{RI}(\text{bY_218}\sim\text{P})]$$

Composition of a Rule-Based Model

a Components



b Interactions



Molecules (4)

```
begin molecule types
Lig(1,1)
Lyn(U,SH2)
Syk(tSH2,1~U~P,a~U~P)
Rec(a,b~U~P,g~U~P)
end molecule types
```

Reaction Rules (19)

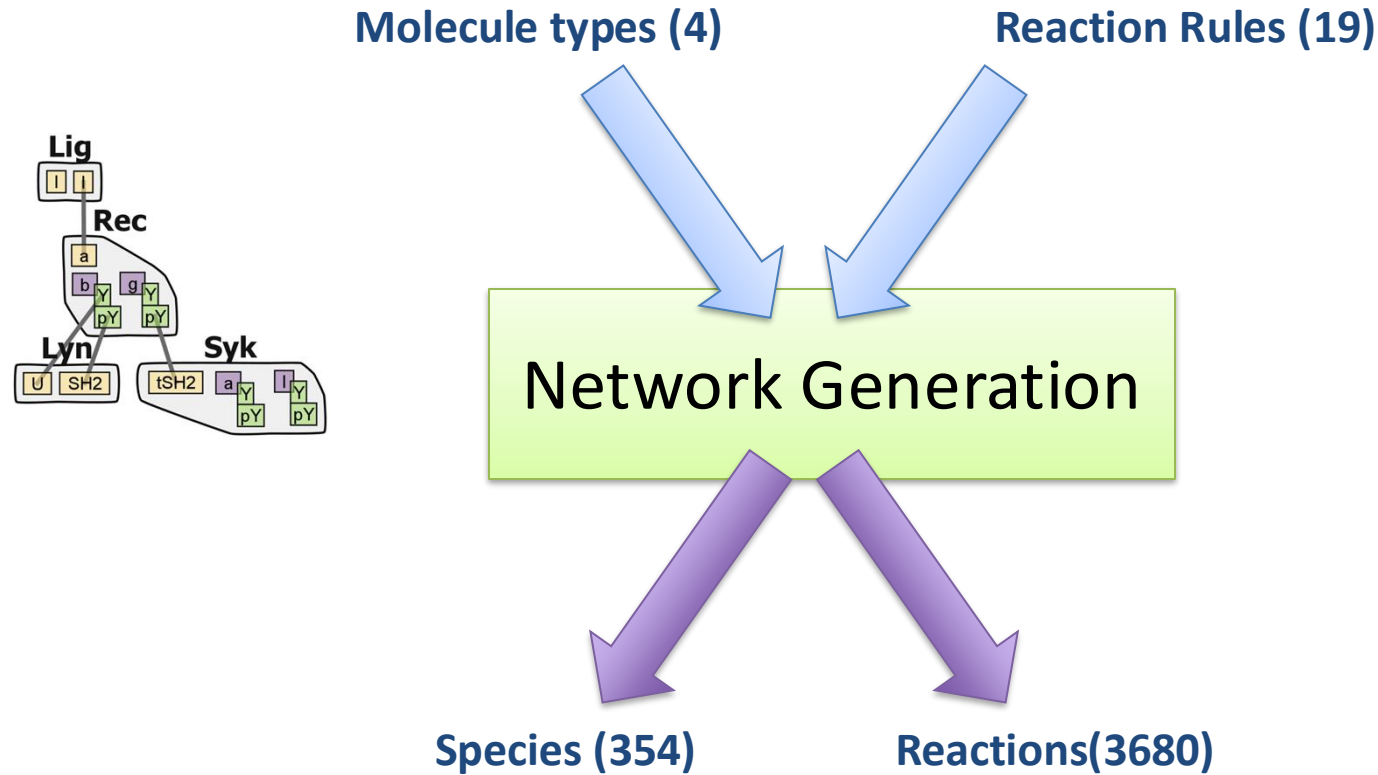
```
begin reaction_rules
# Ligand-receptor binding
1 Rec(a) + Lig(1,1) <-> Rec(a!1).Lig(1!1,1) kp1, km1
  Rec(a) + Lig(1,1) <-> Rec(a!1).Lig(1!1,1) kp1, km1

# Receptor-aggregation
2 Rec(a) + Lig(1,1!1) <-> Rec(a!2).Lig(1!2,1!1) kp2, km2

# Constitutive Lyn-receptor binding
3 Rec(b~Y) + Lyn(U,SH2) <-> Rec(b~Y!1).Lyn(U!1,SH2) kpL, kmL
...
```

BioNetGen language

Composition of a Rule-Based Model



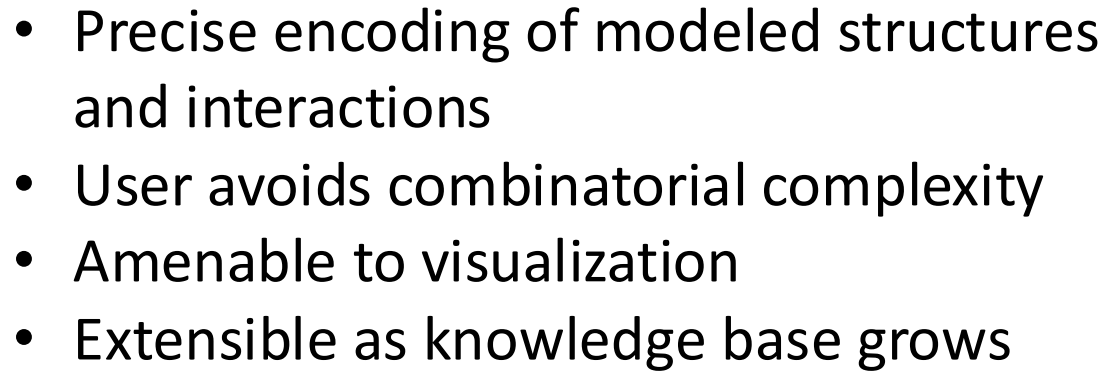
```

1 Lig(l,l) Lig_tot
2 Lyn(SH2,U) Lyn_tot
3 Syk(a~Y,l~Y,tSH2) Syk_tot
4 Rec(a,b~Y,g~Y) Rec_tot
5 Lig(l!1,l).Rec(a!1,b~Y,g~Y) 0
6 Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) 0
7 Lig(l!1,l).Lyn(SH2,U!2).Rec(a!1,b~Y!2,g~Y) 0
8 Lig(l!1,l!2).Rec(a!2,b~Y,g~Y).Rec(a!1,b~Y,g~Y) 0
9 Lig(l!1,l!2).Lyn(SH2,U!3).Rec(a!2,b~Y!3,g~Y).Rec(a!1,b~Y,g~Y) 0
...
    
```

```

1 Rec(a,b~Y,g~Y) + Lig(l,l) -> Lig(l!1,l).Rec(a!1,b~Y,g~Y) 2*kp1
2 Rec(a,b~Y,g~Y) + Lyn(SH2,U) -> Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) kpL
3 Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) + Lig(l,l) -> Lig(l!1,l).Lyn(SH2,U!2).Rec(a!1,b~Y!2,g~Y) 2*kp1
4 Lig(l!1,l).Rec(a!1,b~Y,g~Y) -> Rec(a,b~Y,g~Y) + Lig(l,l) km1
5 Rec(a,b~Y,g~Y) + Lig(l!1,l).Rec(a!1,b~Y,g~Y) -> Lig(l!1,l!2).Rec(a!2,b~Y,g~Y).Rec(a!1,b~Y,g~Y) kp2
6 Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) + Lig(l!1,l).Rec(a!1,b~Y,g~Y) -> Lig(l!1,l!2).Lyn(SH2,U!3).Rec(a!2,b~Y!3,g~Y).Rec(a!1,b~Y,g~Y) kp2
7 Lig(l!1,l).Rec(a!1,b~Y,g~Y) kpL
8 Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) -> Rec(a,b~Y,g~Y) + Lyn(SH2,U) kmL
9 Lig(l!1,l).Lyn(SH2,U!2).Rec(a!1,b~Y!2,g~Y) -> Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) + Lig(l,l) km1
...
    
```

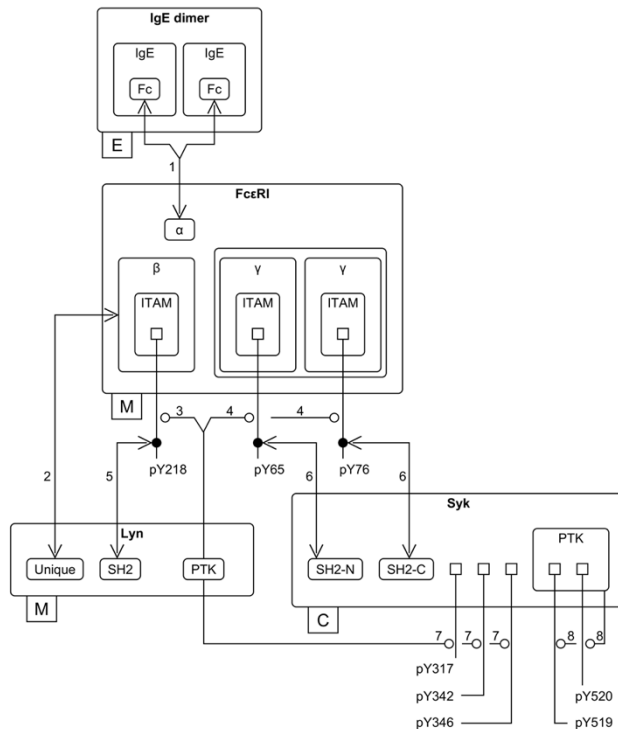
FcεRI model



Faeder et al., *J. Immunol.* (2003)
Chylek et al., *Mol. BioSys.* (2011)

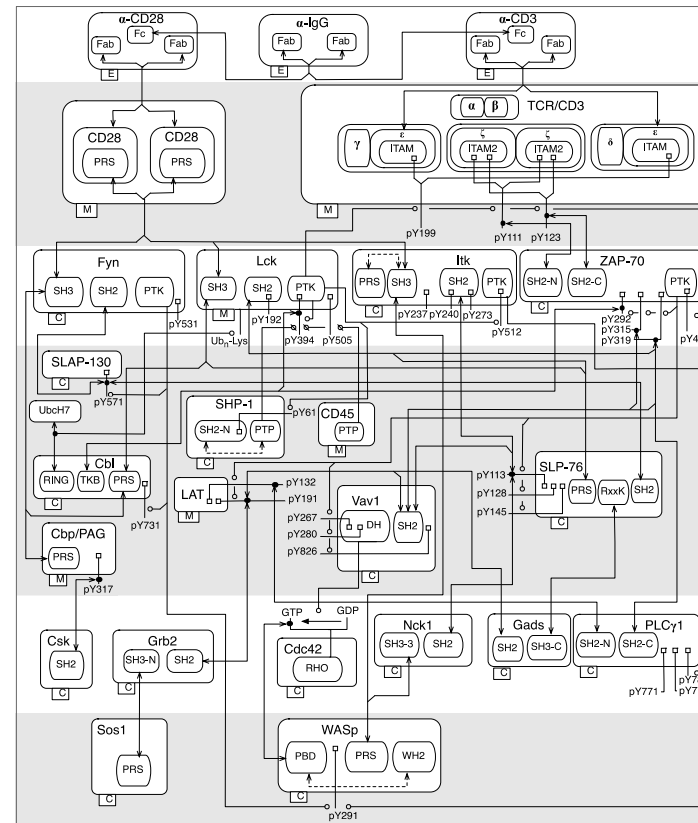
Rule-based modeling enables knowledge representation on a large scale

FcεRI model



Faeder et al., *J. Immunol.* (2003)

T cell receptor model

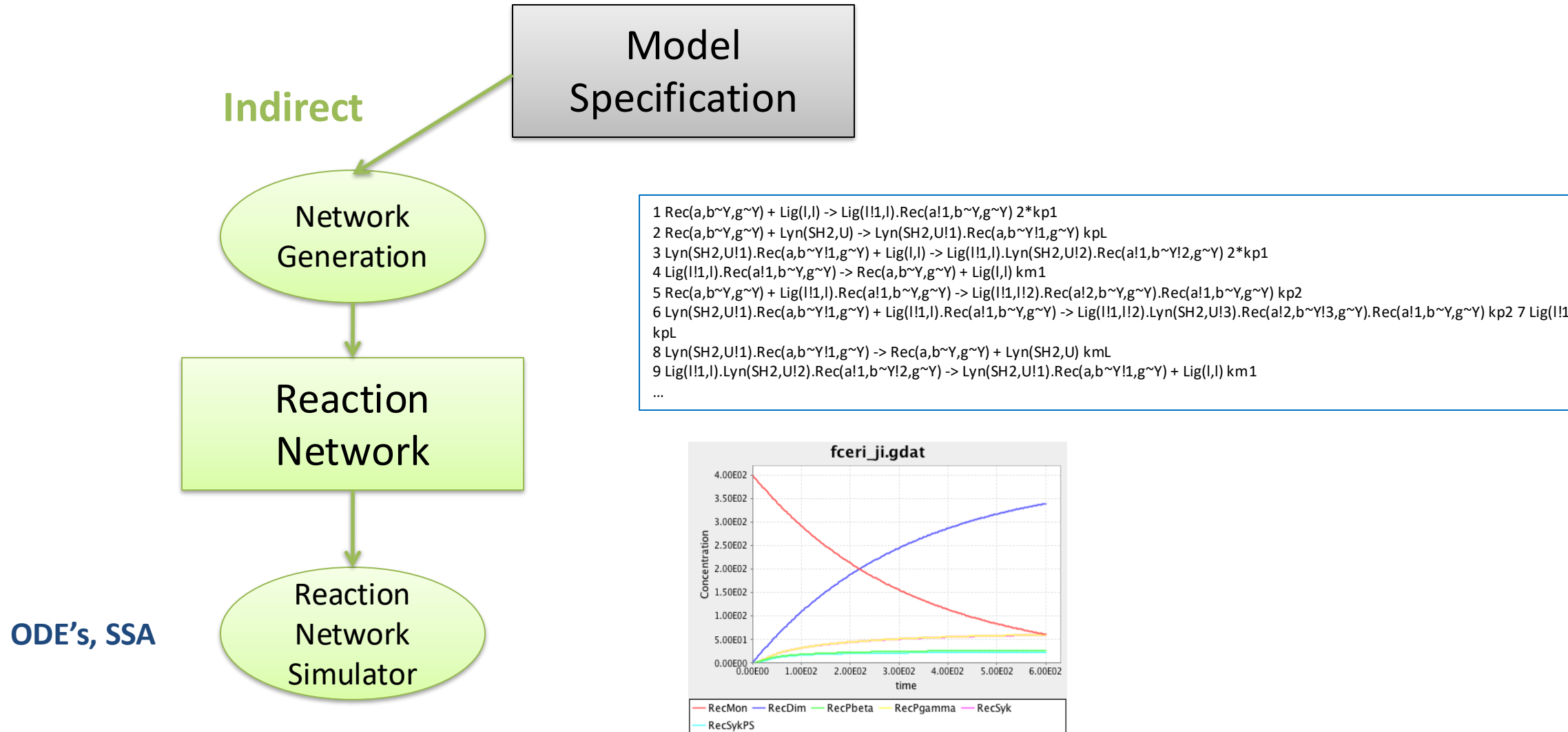


Chylek et al., *PLOS One* (2014)

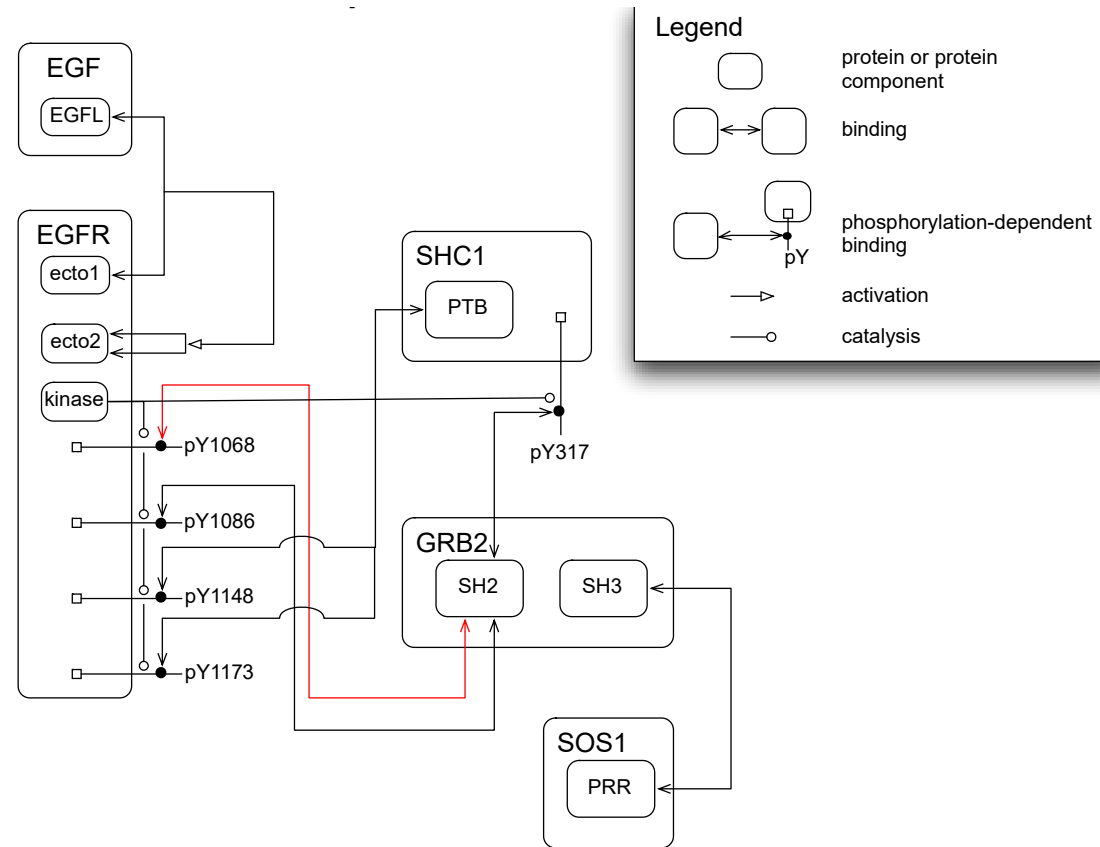
20 molecule types
129 rules

*Number of species is
too large for network
generation!*

Methods for simulating RBMs

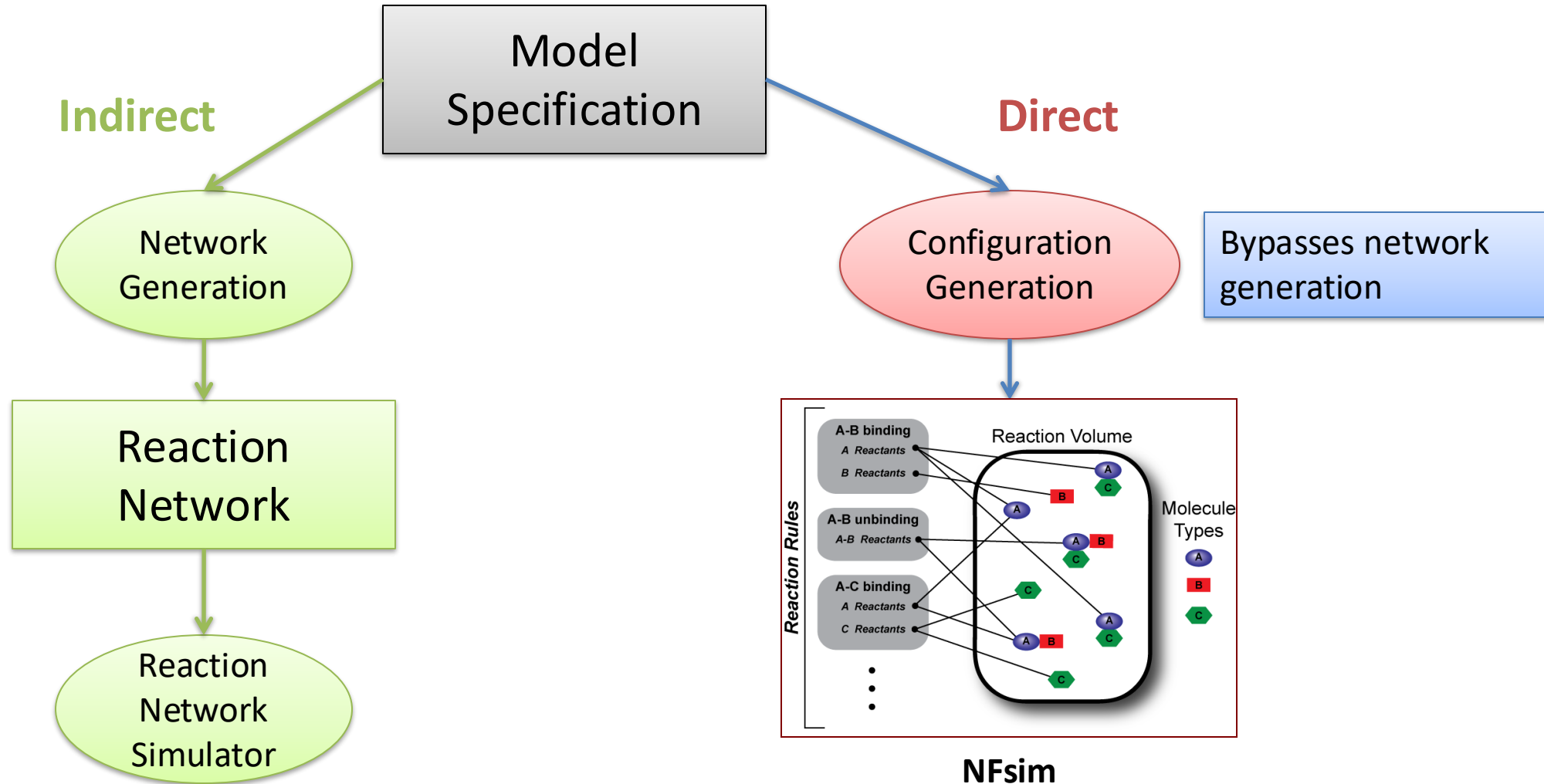


Combinatorial Complexity

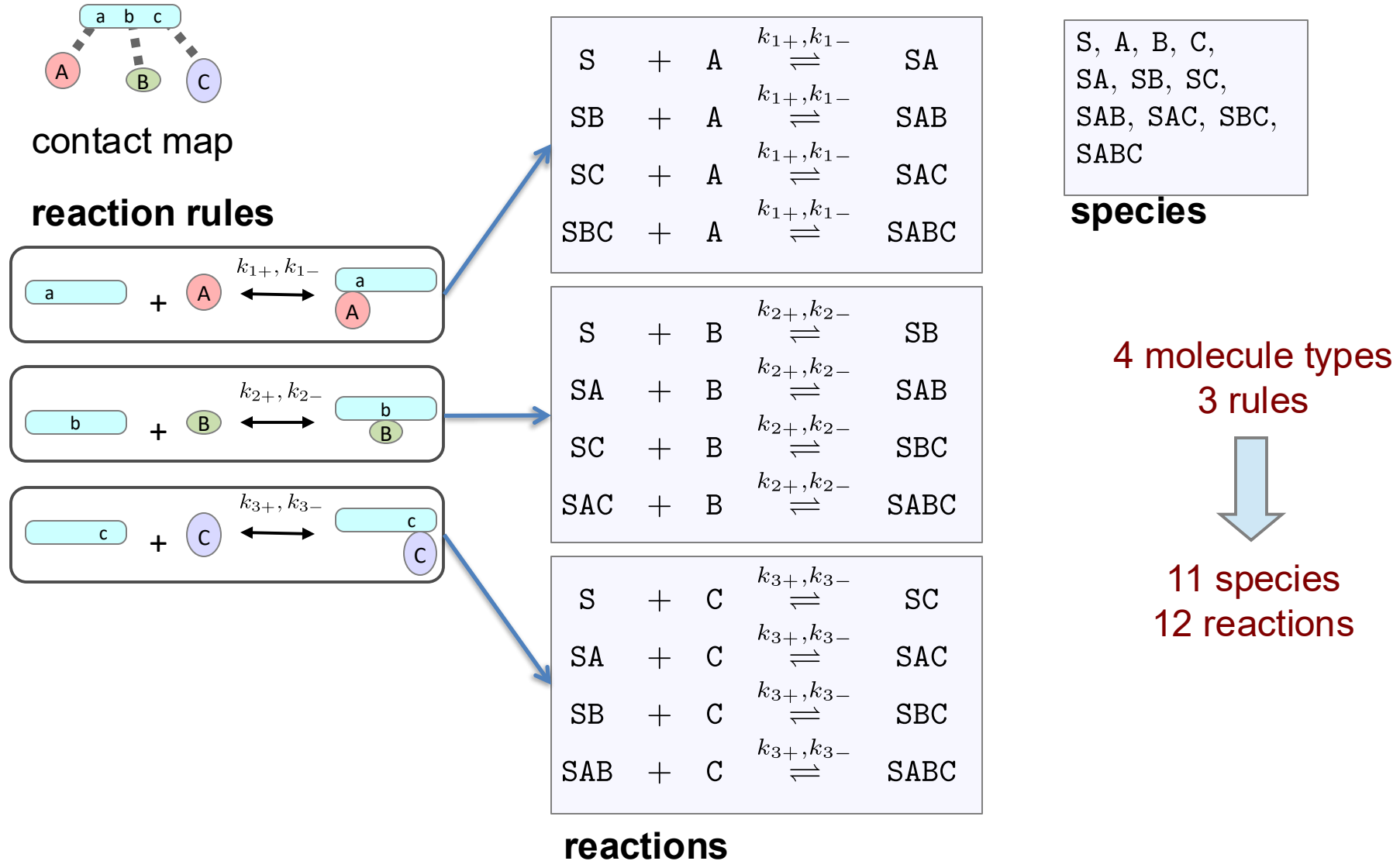


5 proteins, 20 interactions ➡ 170,000 species

Methods for simulating RBMs

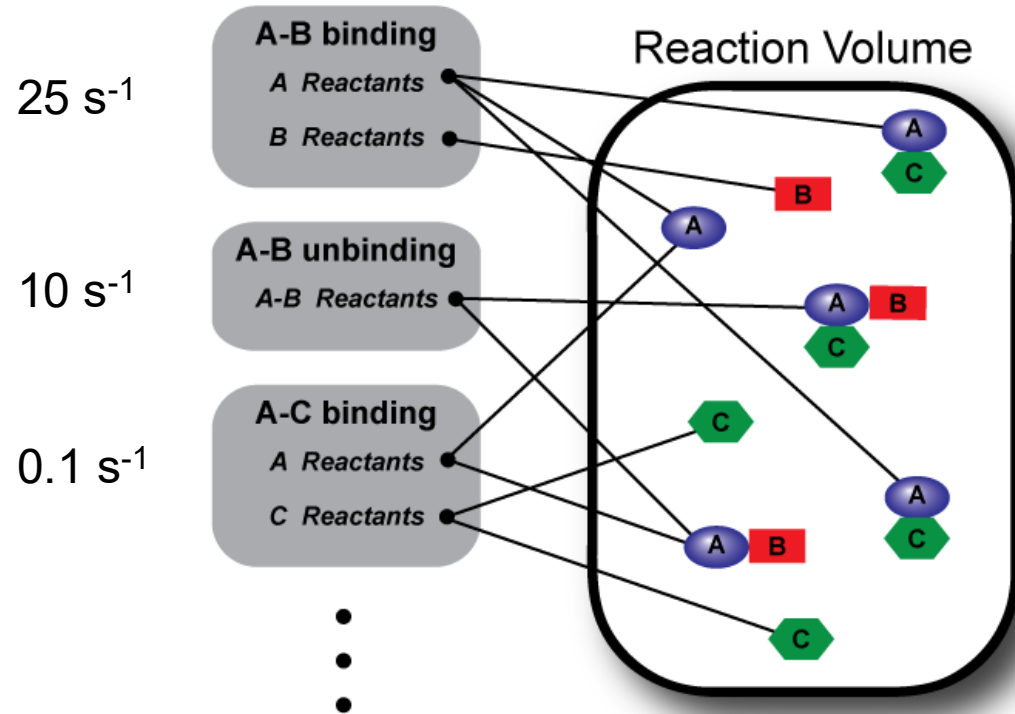


Indirect Methods – Network Generation



NFsim Algorithm

Rates

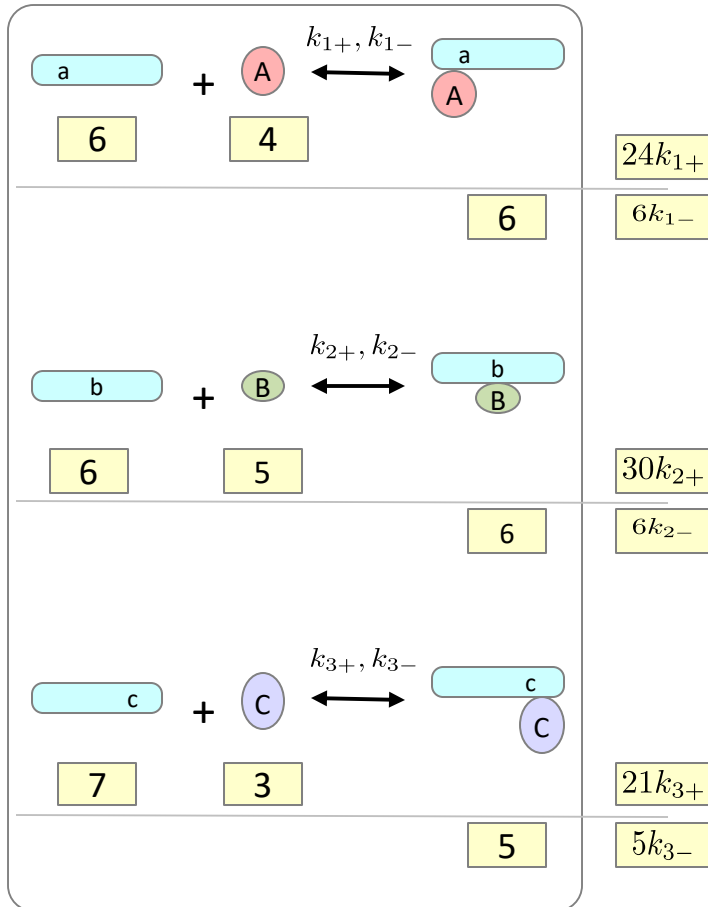


0. Initialize *reactant lists* and calculate *rule propensities*.

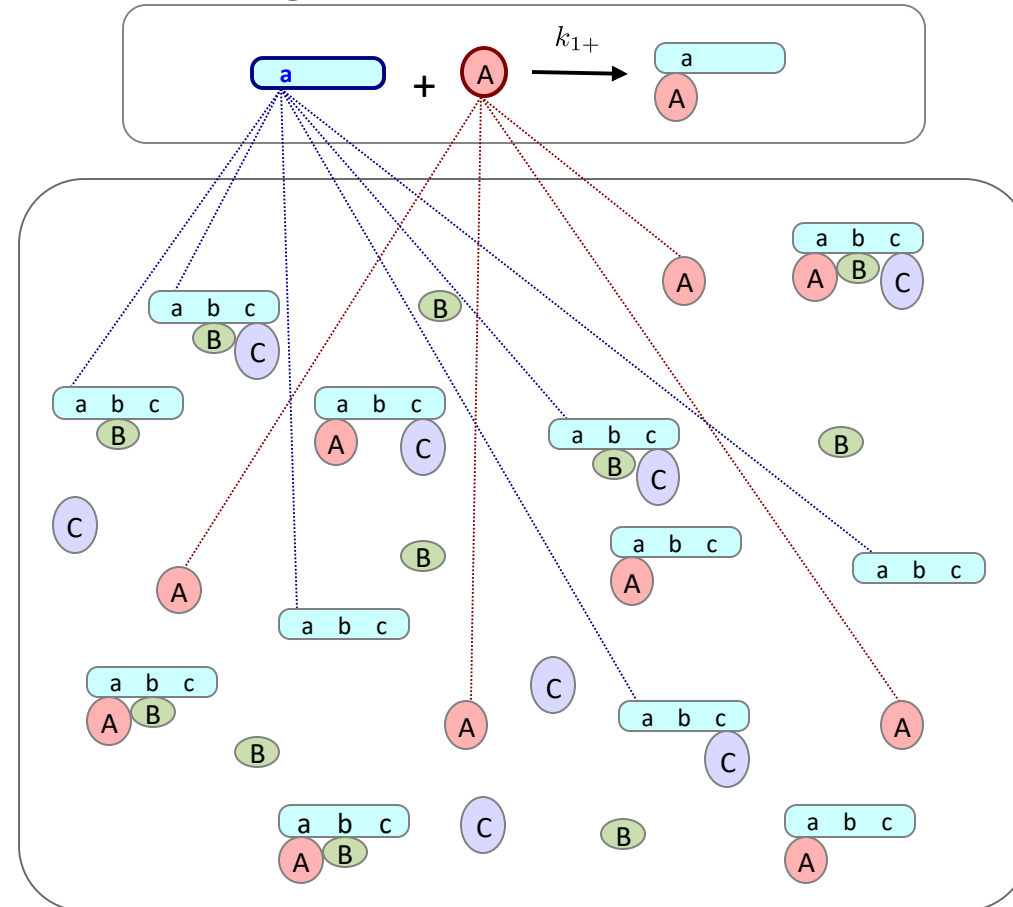
1. Select next reaction time and next *rule*.
2. Select molecules and sites to react.
 - a. Check any application condition(s).
3. Apply operation specified by rule.
4. Update reactant lists and propensities.
5. Increment time.

Direct Methods (NFsim)

reaction rules



event generation



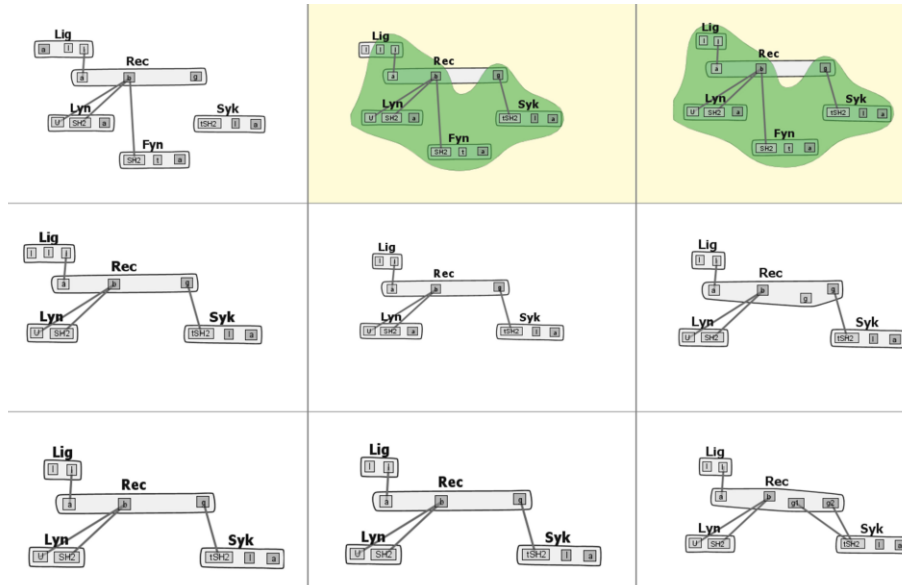
system configuration

total propensity

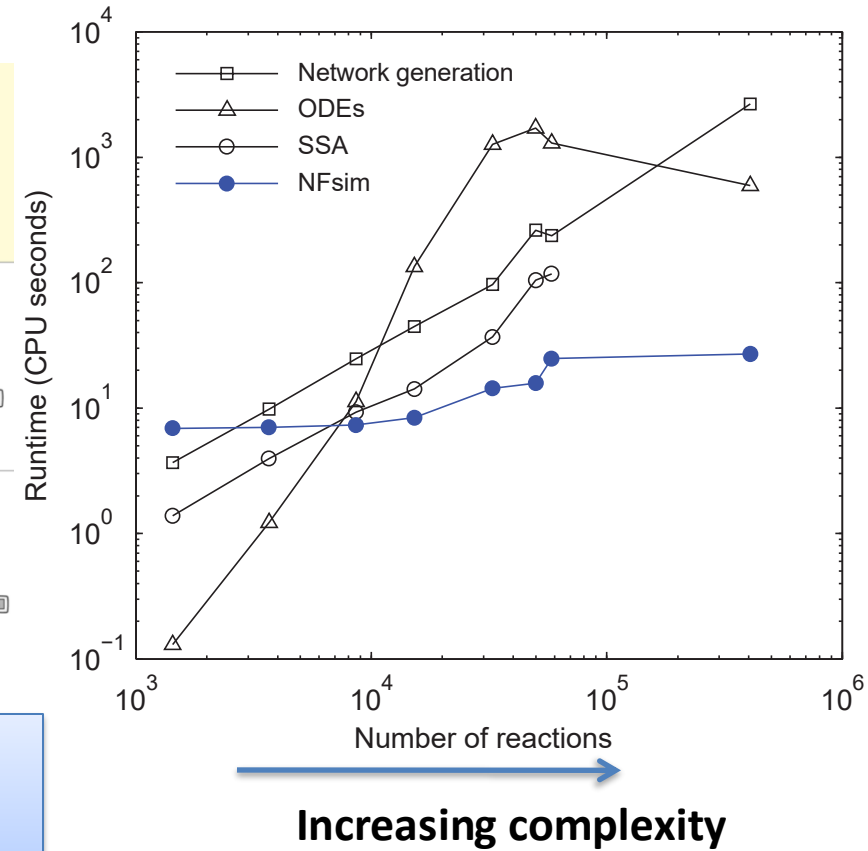
$$24k_{1+} + 6k_{1-} + 30k_{2+} + 6k_{2-} + 21k_{3+} + 5k_{3-}$$

FcεRI signaling models

a



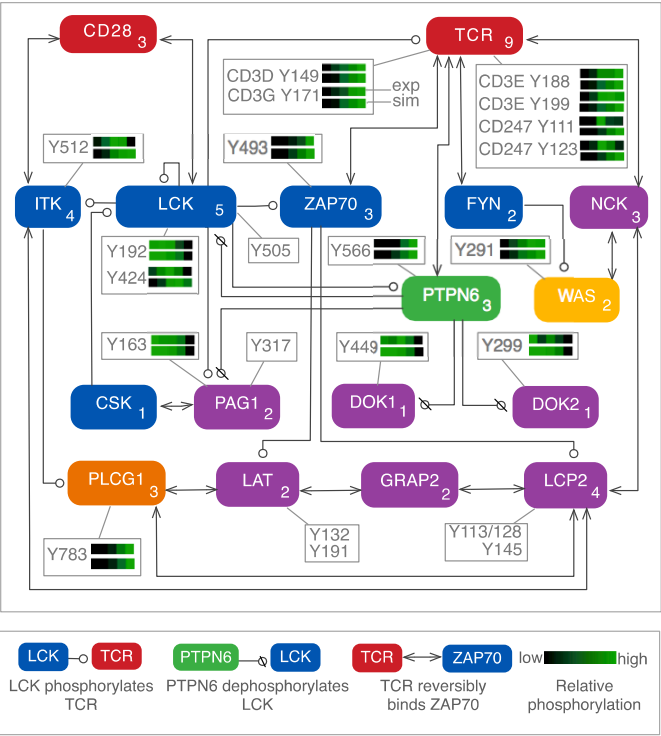
b



NFSIM can simulate models of greatly increased complexity with manageable increase in cost.

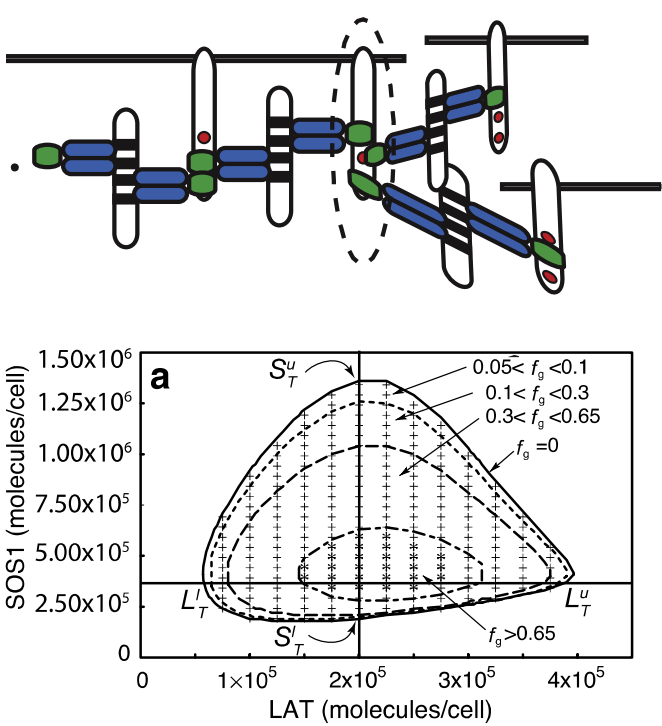
Other examples of network-free applications

Phosphoproteome dynamics



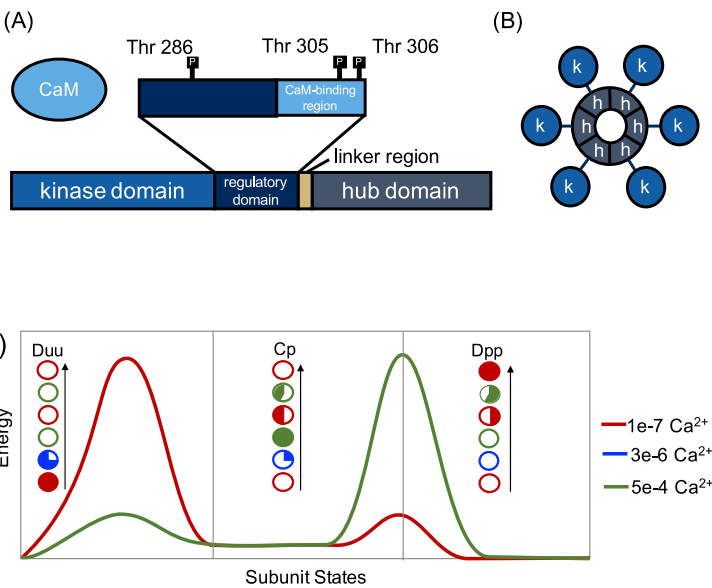
Chylek et al. (2014) *PLOS One*

Multivalent aggregation in signaling



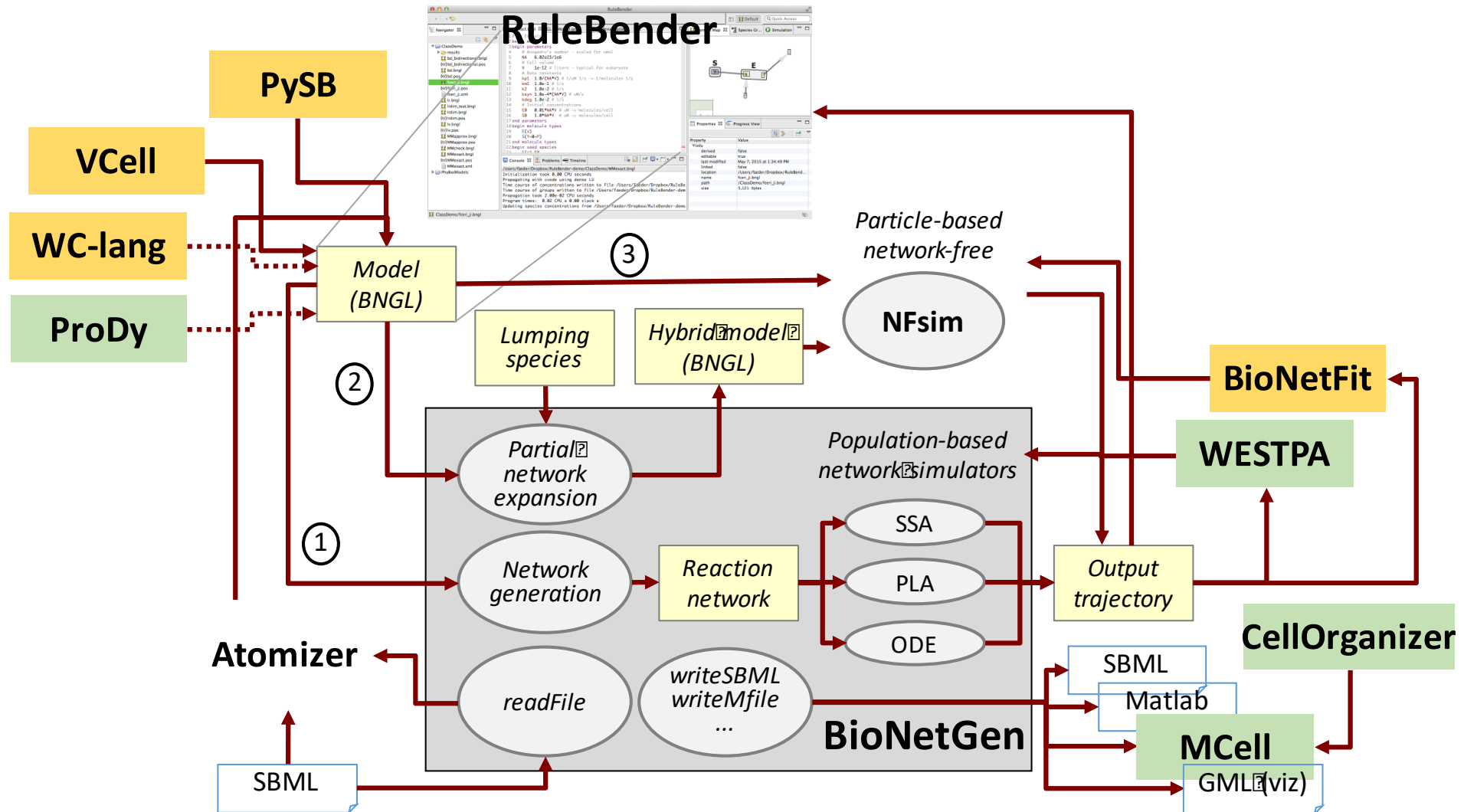
Nag et al. (2009) *Biophys. J*

Holoenzyme dynamics



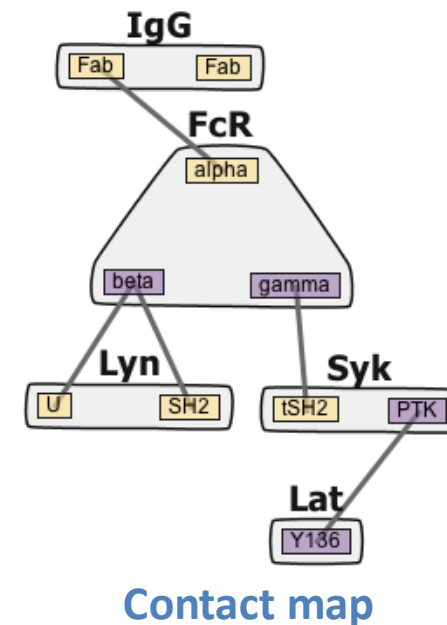
Kaya, Grundy et al., in progress.

BioNetGen and associated modeling tools



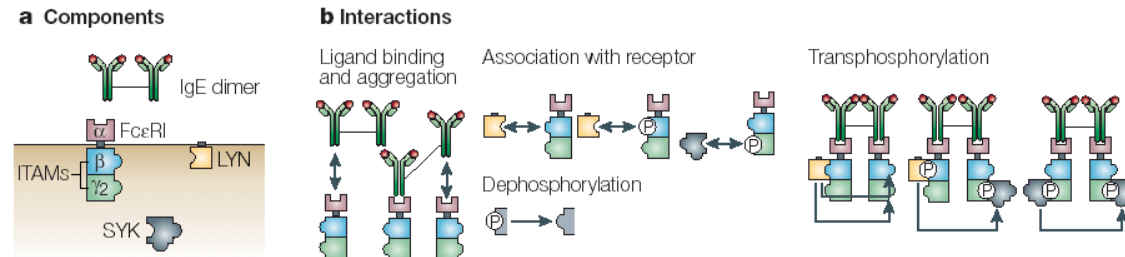
Why use RBM?

- *Concise and precise* representation of biochemical knowledge
 - rules are simple (less context) when interactions are modular
- *Flexible* with respect to simulation method
 - Deterministic / stochastic
 - Well-mixed / compartmental / spatial
- Structures and rules are *reusable*
 - Rule libraries
- Compact visual representation
 - Contact map and beyond

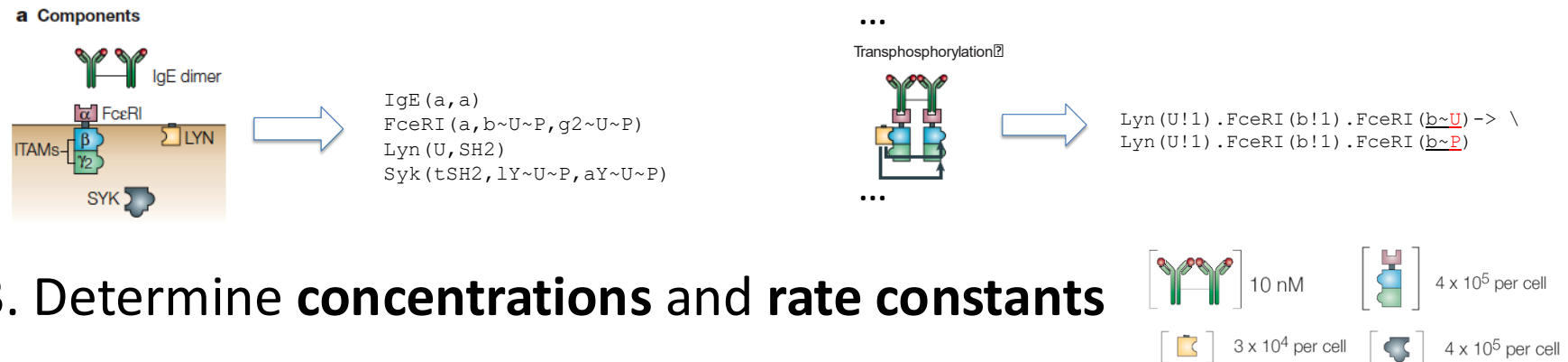


Rule-Based Modeling protocol

1. Identify components and interactions



2. Translate into objects (molecules) and rules



3. Determine **concentrations** and **rate constants**

4. Simulate and analyze the model

ODE's

SSA

PDE's

BD